

BACS 2003
BACS 3074
BMCS 2003
Artificial Intelligence
Tutorial Book

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Please read before answering:

- 1) Ensure you understand the answer you are providing.
- 2) Provide references if necessary.

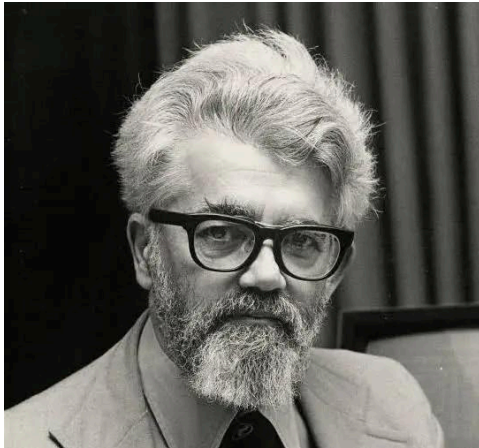
Tutorial 1

1. Who is the father of A.I.? **Describe** the reason why he was recognized so. (Chi Ban)

Answer: John McCarthy,

He is the first person who coined the term "artificial intelligence" and making foundational contributions to the field.Lisp programming....

"



2. Identify **one key event** or major achievement of A.I. development in the year of
- a) 1960-1969 (Ivya)
 - b) 1970-1979 (Han Sheng)
 - c) 1980-1989 (Yan Kang)
 - d) 1990-1999 (Chee Luo)
 - e) 2000-2009 (ZhiQiang)
 - f) 2010-2019 (Jin Zhi)
 - g) 2020-now (Qi Xian)

Answer:

- a) 1960-1969 (Ivya)

- In year 1966, Joseph Weizenbaum designed a programme called ELIZA. The purpose was to stimulate conversation with a human. It used a simple pattern-machine techniques to respond to user inputs, mimicking the behaviour of a psychotherapist. ELIZA did not understand the language truly, but it was a major milestone because it demonstrated that a computer could interact with humans using natural language, which is the foundation for the current modern chatbots and virtual assistants such as Siri and ChatGPT

- b) 1970-1979 (Han Sheng)

- Alain Colmerauer and Philippe Roussel created the programming language PROLOG (Programming in Logic) in 1972. Because PROLOG made it possible for machines to represent and reason using logical relationships, it became one of the most significant languages for AI research, particularly in fields like expert systems and natural language processing.
- c) 1980-1989 (Yan Kang)
 - John Searle created the The Chinese Room thought experiment as a critic towards the Turing Test. The Chinese Room argues that a machine would not be able to have understanding towards knowledge, they can only simulate it, and therefore is not considered as intelligent.
- d) 1990-1999 (Chee Luo)
 - In 1997, IBM's Deep Blue (historic chess-playing computer) made history by defeating world chess champion Garry Kasparov, marking a major milestone in artificial intelligence. This event showed that AI could outperform humans in complex strategic thinking, highlighting significant progress in computer processing power and decision-making algorithms during the 1990s.
- e) 2000-2009 (ZhiQiang)
 - In 2002 First mass produced autonomous robotic vacuum cleaner from iRobot learns to navigate and clean homes
- f) 2010-2019 (Jin Zhi)
 - In 2016 , Google DeepMind's AlphaGo made history by defeating one of the world's top Go player, Lee Sedol. AlphaGo combined deep learning and reinforcement learning to master the game. Its used deep neural networks to evaluate board positions and predict the best moves and also its analyze millions of human games and then by playing against itself millions of times to improve beyond human ability.
- g) 2020-now (Qi Xian)
 - In 2022, OpenAI released ChatGPT, a powerful language model capable of understanding and generating human-like text. This breakthrough marked a major milestone in artificial intelligence, as ChatGPT demonstrated advanced conversational abilities, problem-solving skills, and creativity across various fields.

3. Name an artificial intelligence application that is created by a Malaysian company. Briefly describe its A.I. functions. (Wei Sheng)

Answer:

ILMU is a large-language-model (LLM) type AI system for training on Malaysian context, languages, and culture, developed by YTL AI Labs Sdn Bhd (formerly "FrogAsia Sdn Bhd") in Kuala Lumpur, Malaysia.

Functions:

- It can understand and generate text in multiple languages, including Malay (Bahasa Malaysia)
- Possesses local context-awareness and can provide appropriate responses based on Malaysian culture and everyday situations.
- It supports instruction-following tasks, such as generating content, answering questions, and writing text based on user requests.

4. Differentiate between the following, then provide an example for each type of system.
- a) Systems that think like humans. (Xuan Ping)

- b) Systems that act like humans. (Lik Sheng)
- c) Systems that think rationally. (Zhen Wei)
- d) Systems that act rationally. (Jia Hang)

Answer:

a) (Xuan Ping) - Systems that think like humans try to copy the way people think and learn. They focus on understanding and simulating human thought processes such as decision-making. Example - IBM Watson

b)(Lik Sheng)- **System that behave like human.**



(Fig 1. human that behaves like human)

Example - Sophia the Robot

c) (Zhen Wei) -This approach relates to thinking logically and making decisions based on reasoning/ System that use logic for decision-making, Planning & Optimisation

D) systems that make decisions by choosing the best possible action to achieve their goals based on the information they have.

Example- self driving car (auto drive)

5. Describe input (conditions), processing layer, output and how the agent react for each of the system below:
- i. Agent of an automated class scheduling system. (Shan Yi)
 - ii. Agent of an intelligent air-conditioning system. (Yee Xian)
 - iii. Agent of autonomous driving car. (Juo Sheng)

Answer:

- i. (liew shan yi)

input(condition) - student , lecture , class/timetable

Processing layer - assigning , check availability , avoid clashing

Output - optimised class scheduled

ii) (Yee Xian)

input (conditions) - the system sense room temperature from its sensor

processing layer - rule based logic - apply rule like avoid cooling below 18°C

Output - Adjust temperature, for example if the temperature is more than 26°C, the agent will turn on the cooling.

iii) (Teh Juo Sheng)

Input(condition) - the system senses obstacles/traffic lights colour.

Processing layer - rule based logic - apply rule to avoid collision and moving due to traffic lights colour.

Output - Decide whether to move or stop. Move if green light and also without obstacles infront/Stop if red light and there are objects infront.

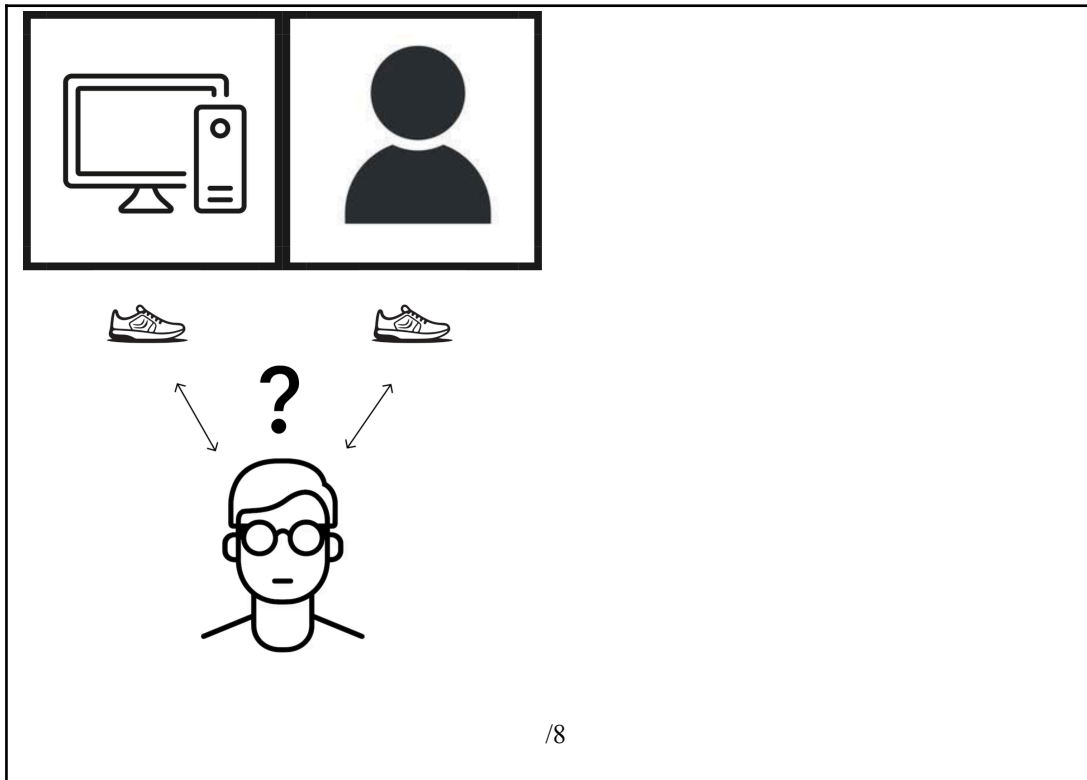
Tutorial 2

Instead of asking, "Can machines think?", Alan Turing said we should ask, "Can machines pass a behavior test for intelligence?". Turing predicted that by the year 2000, a computer could be programmed to have a conversation with a human interrogator for five minutes and would have a 30% chance of deceiving the interrogator that it was a human. (Negnevitsky, 2002).

1. Explain Turing Test. Illustrate your answer with appropriate diagram. (Re Shun)

Answer:

The Turing Test involves 3 roles, a computer, a human, and an evaluator. The evaluator will ask both the human and the computer questions, and receive answers from both. If the computer is able to deceive the evaluator, tricking the evaluator into thinking that the computer is a human, then the machine passes the Turing test.



2. Criticize Turing's criteria for judging a computer's intelligence. (Chi Ban)

Answer:

Focuses on Imitation, Not Intelligence. The Turing Test measures whether a machine can simulate human conversation, not whether it truly *understands* anything.

- The machine uses a huge list of programmed scripts or large memorized databases of conversation to pass the Turing test, but it does no real intelligence or understanding.

3. Suggest how could this test be used (or modified) to assess other kind of artificial intelligence besides a chatbot. Provide an example to elaborate your answer. (Ivya)

Answer:

The Turing test was originally designed to determine whether a machine can exhibit intelligent behaviour indistinguishable from that of a human through conversation. To assess other kind of A.I., the Turing test can be modified to evaluate non-conversational intelligence, such as visual perception, problem-solving or autonomous decision-making.

A modified version of the Turing Test could be applied to self-driving cars. Instead of testing conversation, the test would assess whether an autonomous vehicle can make driving decisions (like braking, changing lanes, and following traffic rules) in a way indistinguishable from a human driver.

4. The Loebner Prize is an annual competition in artificial intelligence that awards those computer programs considered by the judges to be the most human-like, using format of a standard Turing Test. The conversation scope between the programs and the judges has been unrestricted since 1995, and the duration of the conversation has been increased from 5 minutes to 25 minutes since 2010 (<http://www.loebner.net/>). (Qi Xian)
- Discuss **TWO (2)** reasons why Turing Test is considered **not effective enough** in assessing machine intelligence.
 - Discuss **TWO (2)** challenges to build a computer program that can win the Grand Loebner Prize, in which judges totally cannot distinguish it from a real human.

Answer:

- i) Focuses on imitation, not true understanding
 - The program is more like mimicking human reactions than truly understanding their meaning or context.
 - It measures how “human-like” the conversation sounds, not whether the machine truly thinks or reasons.
 - ii) Subjective human judgment
 - Different judges may have different expectations of what sounds human
- i) need large amount of dataset
 - ii) Avoiding mechanical or repetitive responses
 - Repetition or unnatural phrasing quickly reveals a non-human speaker.
 - The program needs to generate diverse, context-appropriate sentences to sound genuinely human

5. The Chinese room argument by John Searle is one of the best known and widely credited criticism of Turing Test. Explain John Searle's Chinese room concept. (Zhen Wei)

Answer:

- Chinese room argument concept is based on reaching out unknown answer by comparing the rules given. It criticizes that machine was not ‘intelligence’ based on only choosing the pictures during Turing tests.

6. Try to chat with the following chatbots within a few minutes. Then discuss what are the characteristics/behaviors of a chatbot should have in order to deceive any human. (Yee Xian)
- Mitsuku**, the 5-time Loebner Prize winner - <https://www.pandorabots.com/mitsuku/>
 - Eliza**, the first chatbot - <https://web.njit.edu/~ronkowitz/eliza.html> (not the original Eliza website)

Answer:

- a)- Use slang during conversation to match user's style
 - Joke or compliment user during conversation

- b) -Use punctuation marks like the exclamation mark.and question mark
 - Refuse to answer some question
 - Ask question about user

Tutorial 3

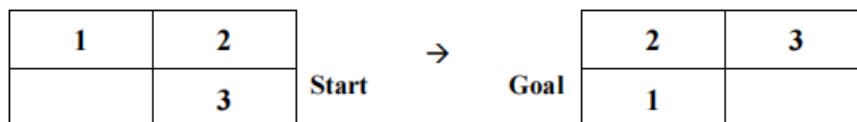
1. Discuss the advantages and disadvantages of **breadth-first search** and **depth-first search**. (Qi Xian)

Answer:

	breadth-first search	depth-first search
Advantages	Complete – It will eventually explore all nodes level by level until the goal is found.	Fast when the solution(goal) is deep – Goes straight down to depth before exploring others high level nodes
Disadvantages	Slow when the solution(goal) is deep – need to explore unnecessary nodes in higher levels first.	Does not guarantee the shortest path – might go deep into the wrong direction.

2. **Figure 1** shows a puzzle problem that requires rearrangement of the tiles to transform the order from start to goal state. One is only permitted to slide the empty tile follow the sequence of **left**, **right**, **up** or **down**.

Remark: Avoid repeated state



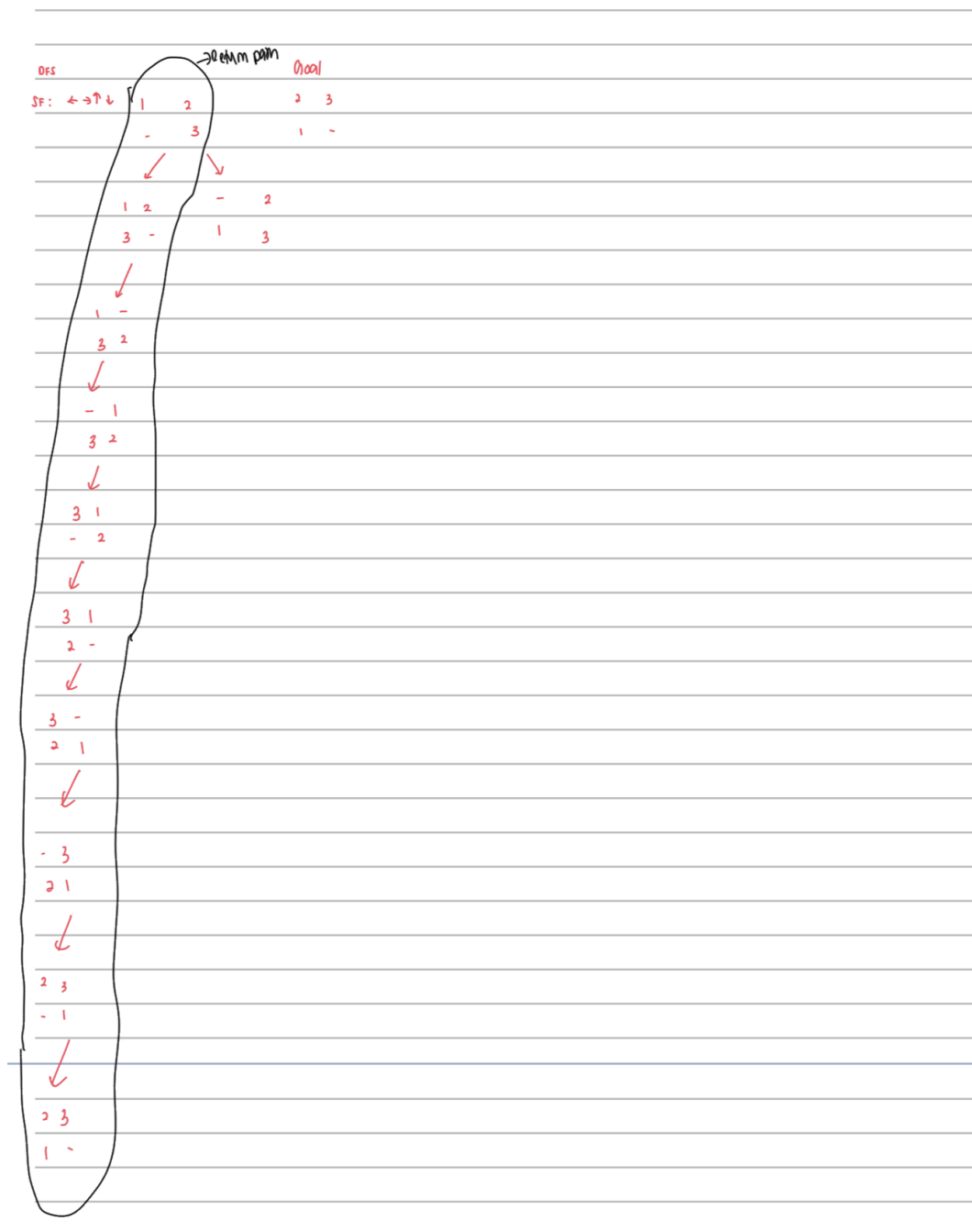


Figure 1: The Puzzle Problem

- Provide the goal formulation of the puzzle problem above. (Jin Zhi)
- Formulate the puzzle problem above by specifying the initial state, successor functions, goal test, step cost, and path cost. (Jia Hang)
- Perform **breadth-first search** and **depth-first search** on the puzzle problem above. Draw the resulting search trees for both. (Ivya)

1a) goal: 2 3

1

Optimal Solution: to achieve the goal in shortest path

Abstraction: time

2b) initial state : $[1, 2, _, 3]/$

Successor functions : move the empty tile according to the sequence of left ,right, up , down

Goal test : a function to check if current state == [2, 3, 1, _]

Step cost : cost between 2 stations, in this case, always be 1

Path cost : function to calculate total step cost along a path

3. **Figure 2** shows a puzzle problem that requires rearrangement of the tiles to transform the order from start to goal state. One is only permitted to slide the empty tile follow the sequence of **down, right, up or left**

Remark: Avoid repeated state

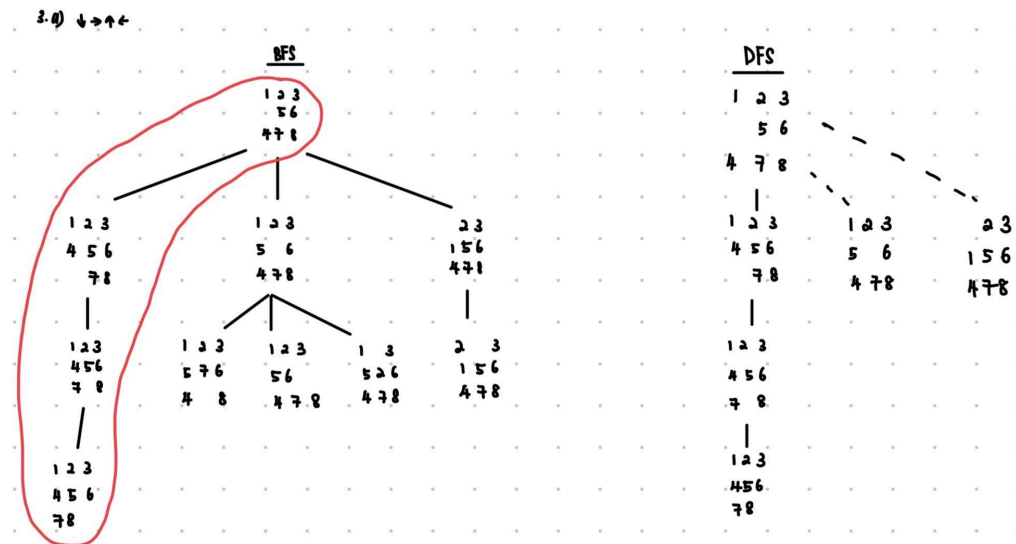
1	2	3			1	2	3
	5	6			4	5	6
4	7	8	Start	→	7	8	Goal

Figure 2: The 8-Puzzle Problem

- Perform **breadth-first search** and **depth-first search** on the 8-puzzle problem above. Draw the resulting search trees for both. (Chee Luo)
- Evaluate the efficiency of **breadth-first search** and **depth-first search** in terms of completeness, optimality, time efficiency and space efficiency in solving the problem above. (Chi ban)

Answer:

a)



b)

	BFS	DFS
completeness	Yes	Yes
Optimality	Optimal	optimal.

Time efficient	Low	high
Space efficiency	low	High

4. In the family river-crossing problem, two parents are with their two children - a son and a daughter - came to a wide river. The only way to get to the other side was to ask a fisherman if he could lend them his boat. However, the boat could carry only two persons. For safety reason, no child should be left alone without the supervision of at least one parent. The family must get to the other side and finally returns the boat to the fisherman, assuming only the fisherman and the two parents know how to row the boat. Based on this family river-crossing problem, answer the following questions.

Remark: Avoid repeated state

- Suggest a simple representation of the initial state. You must briefly explain the representation. (Zhi Qiang)
- Describe the goal formulation and problem formulation: initial state, successor function, step cost, path cost and goal test. (Wei Sheng)
- Draw the depth-first search tree to show how all the states are being traversed. (Yan Kang)

Answer:

a) [2P, 2C, 1F, L, 0P, 0C, 0F]

2P: 2parents

2c: 2 Child

1F: 1 fisherman

L: boat position at left

R: boat position at right

b) initial state: [2P, 2C, 1F, L, 0P, 0C, 0F]

successor function: 1F, 1F1P, 1P1C, 1P go to right side or return

step cost: 1

path cost: 8

[1P, 2C, 0F, R, 1P, 0C, 1F]

[1P, 2C, 1F, L, 1P, 0C, 0F]

[1P, 1C, 0F, R, 1P, 1C, 1F]

[1P, 1C, 1F, L, 1P, 1C, 0F]

[1P, 0C, 0F, R, 1P, 2C, 1F]

[1P, 0C, 1F, L, 1P, 2C, 0F]

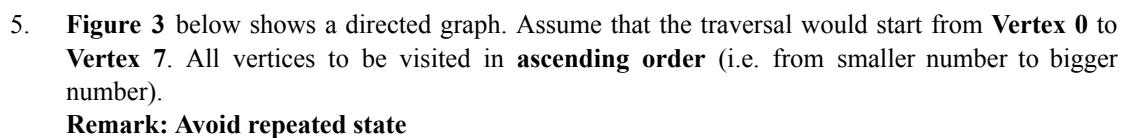
[0P, 0C, 0F, R, 2P, 2C, 1F]

[0P, 0C, 1F, L, 2P, 2C, 0F]

goal test: All family members are on the right side, and return the boat to the fisherman.

[0P, 0C, 1F, L, 2P, 2C, 0F]

c)




 (:-<<<<<<<<<

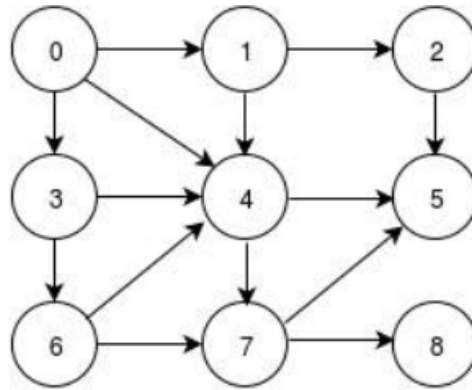
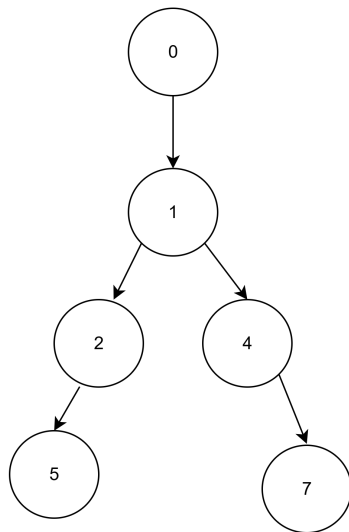


Figure 3: The Directed Graph

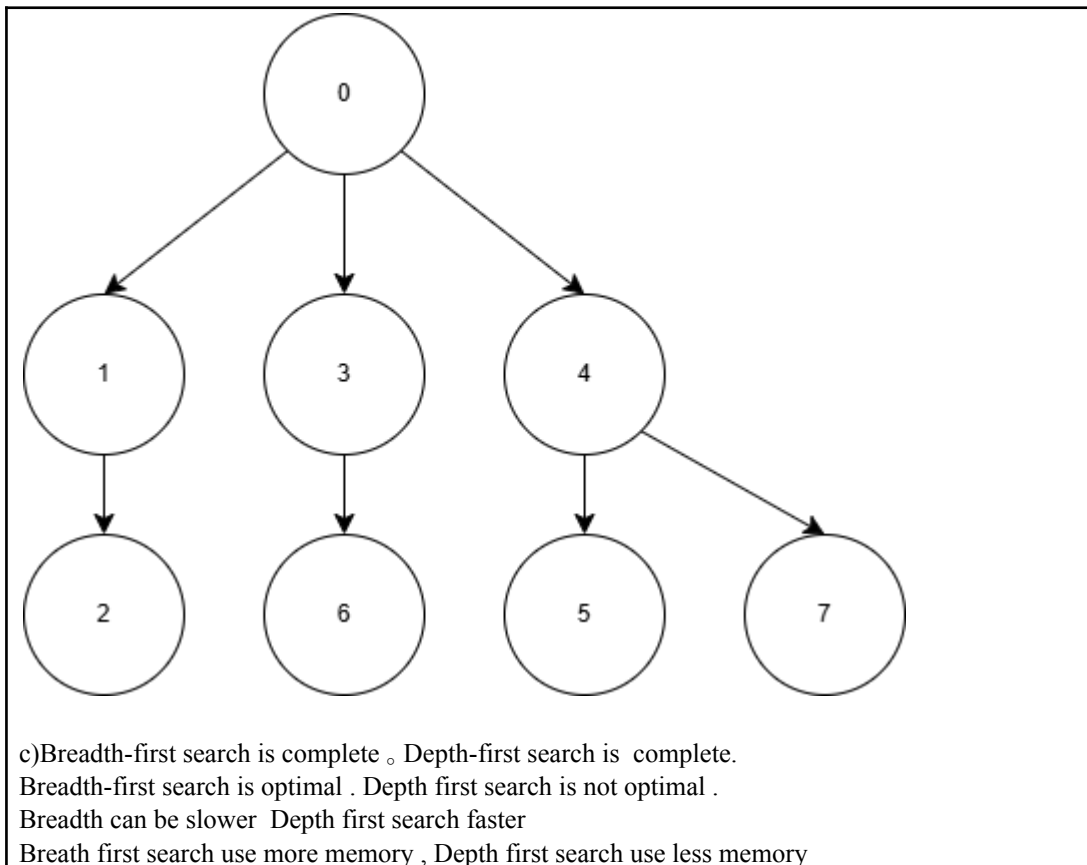
- Perform a **depth-first search** on the directed graph to traverse from Vertex 0 to Vertex 7. Draw the resulting search tree and list the returned path. (Han Sheng)
- Perform a **breadth-first search** on the directed graph to traverse from Vertex 0 to Vertex 7. Draw the resulting search tree and list the returned path. Avoid repeated state. (Juo Sheng)
- Evaluate the efficiency of **breadth-first search** and **depth-first search** in terms of completeness, optimality, time efficiency and space efficiency in solving the problem above. (Lik Sheng)

Answer:

- a) Return path: 0, 1, 4, 7



- b) Returned path: 0 > 4 > 7



Tutorial 4

1. **Figure 1.1** below shows a Block World Problem. A robot will move the blocks one by one from initial state S1 to reach the goal state S19.

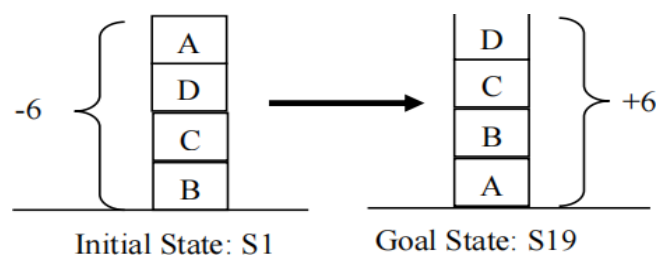


Figure 1.1

Figure 1.2 shows the state space of the Block World Problem and the heuristic costs for each state are shown in parentheses next to their respective nodes.

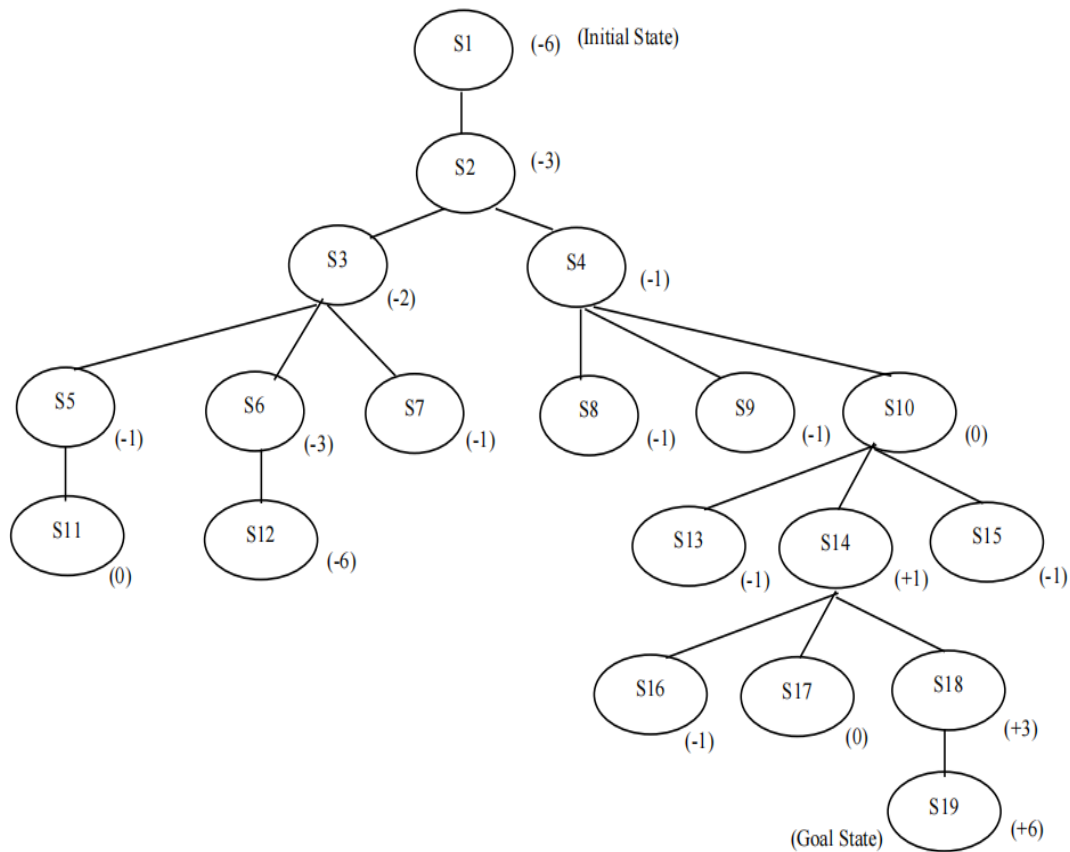


Figure 1.2

- Explain step cost used in problem formulation. Specify the value of the step cost for the problem above. (Yeexian)
- Hill climbing search is unable to guarantee completeness and optimality as it may be trapped into local maximum. (Zhen Wei)
 - Explain local maximum.
 - Discuss why hill climbing search always lead to a local maximum.

Use simple hill-climbing and steepest-ascent hill-climbing to search for the best path from S1 to S19 on the state space shown in **Figure 1.2**. Then for each search technique, draw the resulting search tree that shows the visited nodes. Show that hill-climbing technique can be trapped into a local maximum. (**Remark: Avoid repeated state**)
- A search technique can be evaluated based on four criteria: completeness, optimality, time complexity and space complexity. Evaluate the efficiency of breadth-first search and steepest-ascent hill-climbing. Conclude which technique is better to solve the Block World Problem mentioned in **Figure 1.1**.
(Remarks: you are required to perform **breadth-first search before comparing**) (Re Shun)

Answer:

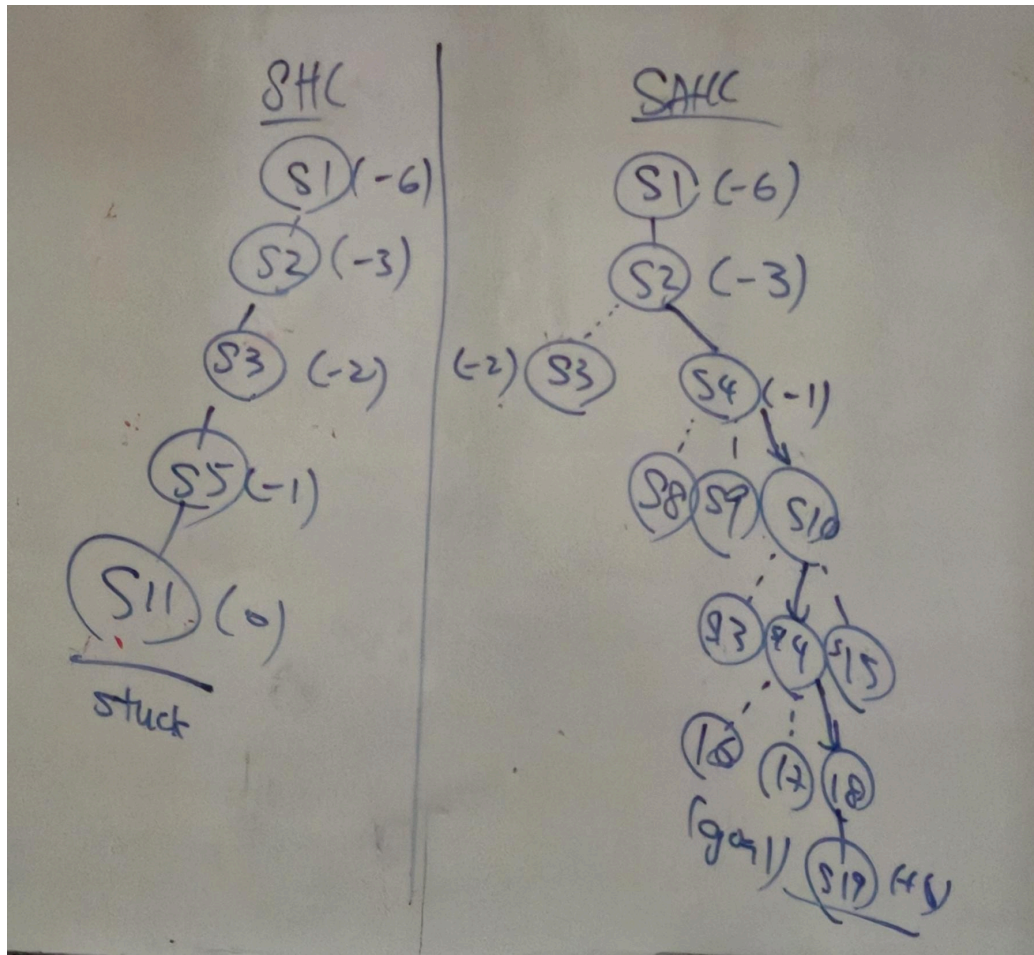
a) Step cost represents the cost of performing a single action or moving from one state to another state in the search space. Based on Figure 1.2, the step cost value for each transition in the Block World Problem is 1, since the robot only moves one block at a time and all moves have equal cost

b) i) A local maximum is a point in the search space where the current solution is better than all of its immediate neighbours, but is not the best possible solution overall. In other words, it is a “peak” that is higher than nearby points, but lower than the global maximum. A search algorithm can get stuck here because every neighbouring move appears worse, even though a much higher peak exists elsewhere.

b) ii) Hill climbing is a greedy search algorithm that always chooses the best neighbouring state according to a heuristic (i.e., it moves “uphill” in value). Because:

- It only considers immediate neighbours of the current state.
- It accepts only improving moves.
- It never backtracks or explores worse states that could lead to a higher peak later.

As a result, when it reaches a state where no neighbour is better, the search stops.



c)

Breadth-First Search:

Return path: S1 > S2 > S4 > S10 > S14 > S18 > S19

Steepest-Ascent Hill-Climbing:

Return path: S1 > S2 > S4 > S10 > S14 > S18 > S19

	Breadth-First Search	Steepest-Ascent Hill-Climbing
Completeness	Complete	Complete
Optimality	Optimal	Optimal
Time Complexity	High	Low

Space Complexity	High	Low
Conclusion: Steepest-Ascent Hill-Climbing is the better technique to solve the Block World Problem mentioned in Figure 1.1 .		

2. **Figure 2** below shows an 8-puzzle problem, which requires the rearrangement of the tiles to transform the order from the start state to the goal state. One is only permitted to slide the empty tile in the sequence of **up, down, right or left**. (Remark: Avoid repeated state)

1	2	3	→	1	2	3
	4	5		4	5	6
7	8	6		7	8	
Start				Goal		

Figure 2: The 8-Puzzle Problem

- Suggest a heuristic function to produce a heuristic cost for a state. Demonstrate how such heuristic cost can be computed on the **start state**. Then perform best-first search. (chi ban)
- Evaluate the efficiency of **breadth-first search** and **best-first search** in terms of completeness, optimality, time efficiency and space efficiency in solving the problem above. (Remarks: You are required to perform **breadth-first search before comparing**) (Juo Sheng)
- Can hill-climbing find a solution for this problem? Draw a resulting search tree to support your answer. (Wei Sheng)

Answer:

2(a) $h(n)$ = tile at the correct position.

Date. . .

2) a) $\uparrow \downarrow \leftarrow \rightarrow$

10

1 2 3

4 5 ($h=5$)

7 8 6

10

2 3

1 2 3

1 2 3

($h=4$) 1 4 5

7 4 5 ($h=4$)

4 5 ($h=6$)

$\leftarrow$ 5 7 8 6

8 6 $\leftarrow$ 5

7 8 6

($h=5$)

($h=5$)

($h=7$)

1 3

1 2 3

1 2 3

4 2 5

4 8 5

4 5

7 8 6

7 6

7 8 6

($h=6$)

($h=8$)

1 2

1 2 3

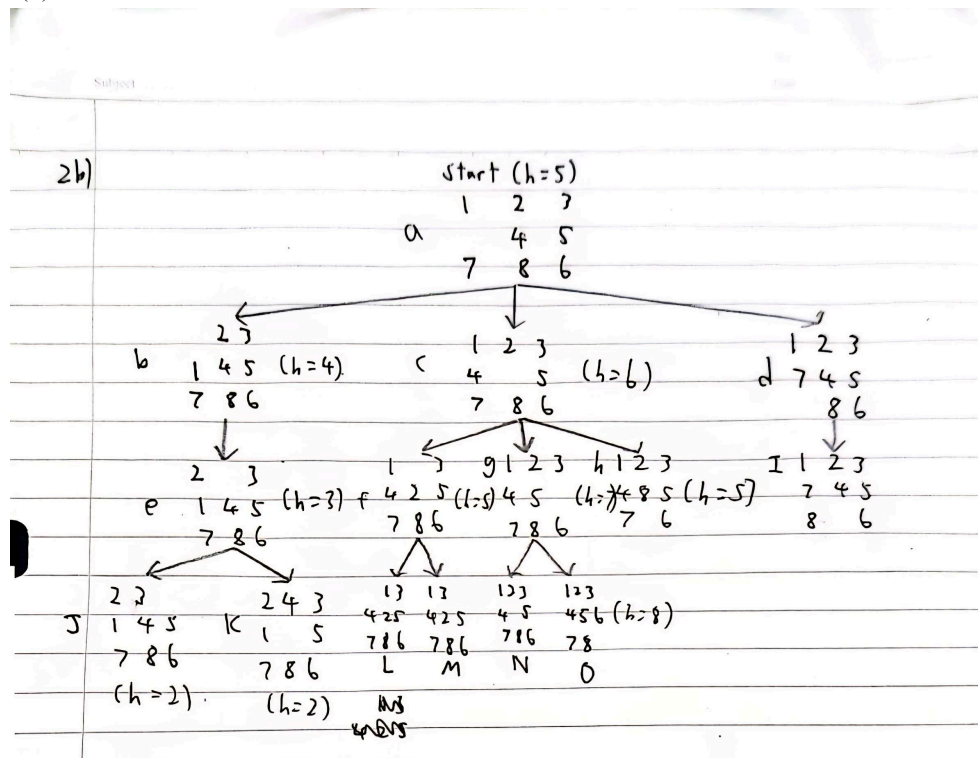
4 5 3

4 5 6

7 8 6

7 8

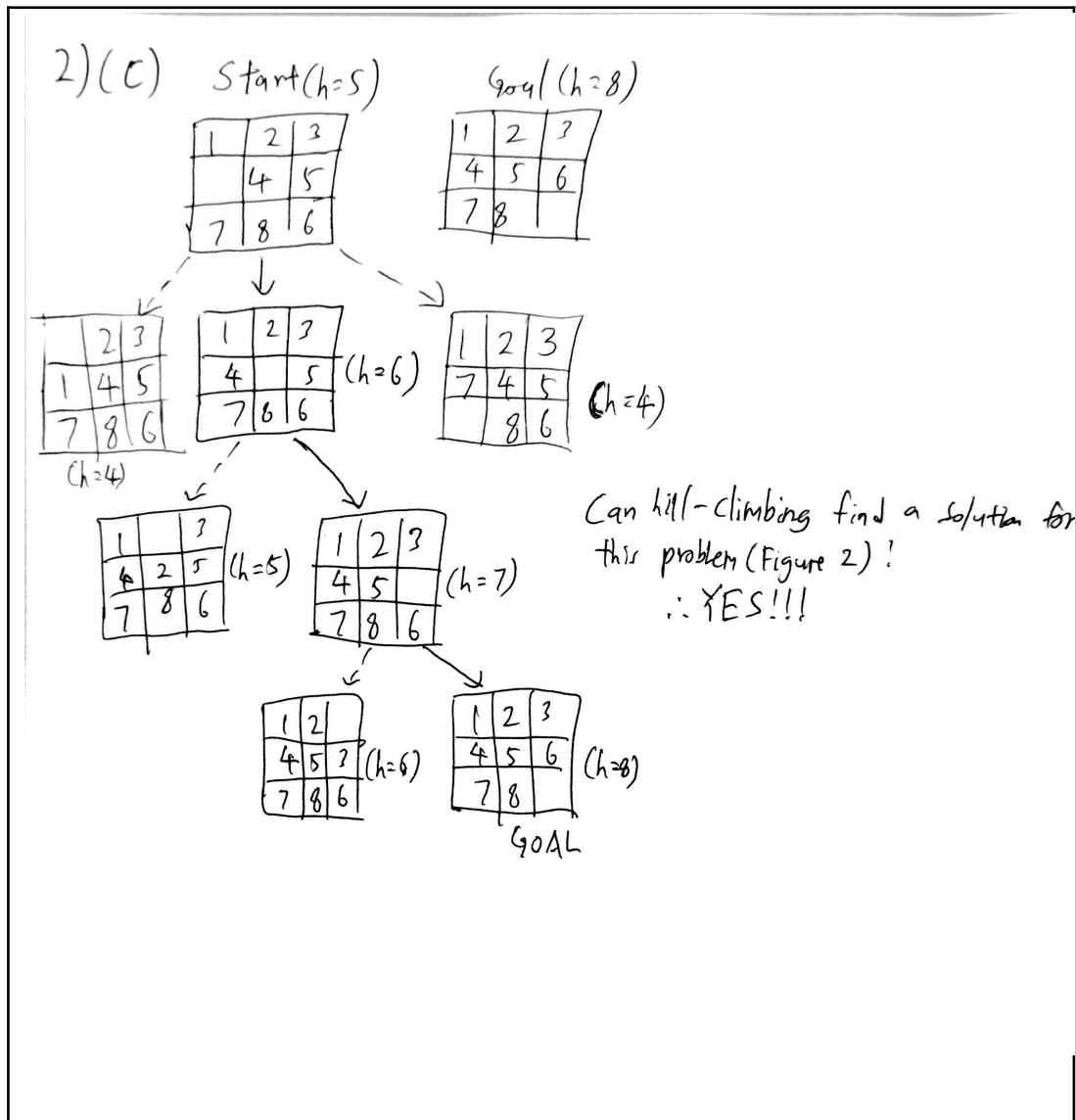
• 2(b)



• Return Path a>c>g>o

	Breadth-First Search	Best-First Search
Completeness	Complete	Complete
Optimality	Optimal	Optimal
Time Efficiency	Low	High
Space Efficiency	Low	High

2(c)SHC



3. The following graph in Figure 1.1 shows all the nodes in a telecommunication network. The distance (in km) from one node to another is shown on the arc.

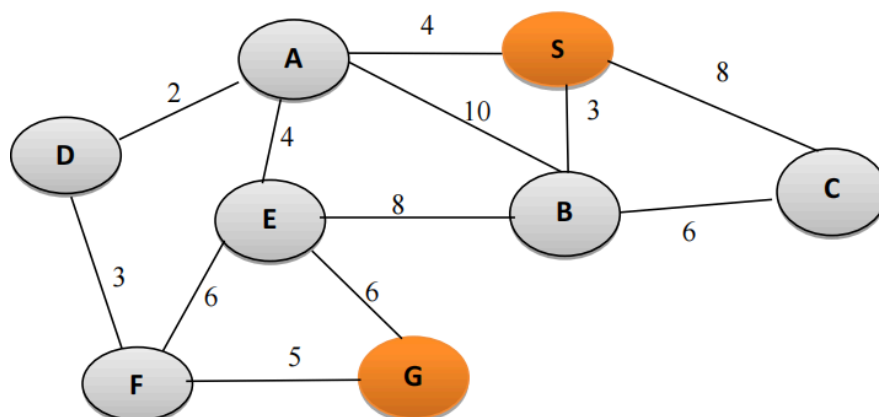


Figure 3: A search graph of a new LRT network

The Euclidean distance (in km), which is used as the heuristic cost (h) for different nodes, is provided in Table 1 below.

S	A	B	C	D	E	F	G
14	10	13	8	8	11	5	0

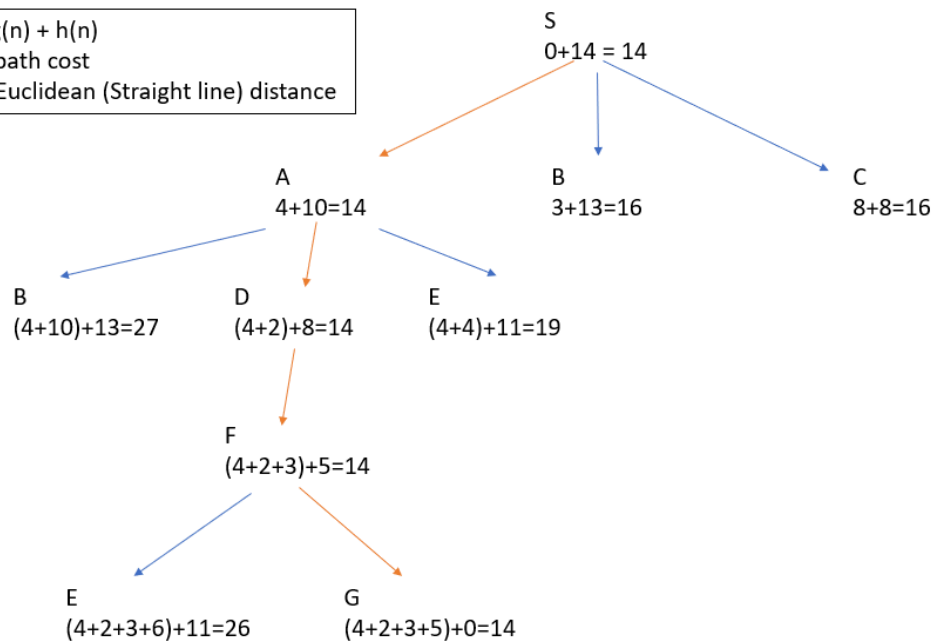
- Assume that some data are to be sent from node S to node G using the shortest route. Describe the goal formulation and problem formulation (initial state, successor function, goal test, step cost and path cost).(Shan Yi)
- Show the resulting search tree of A* search to find the shortest path from S to G. State the shortest path. (**Remark: Avoid repeated state**)(Yan Kang)
- Evaluate the efficiency of A* search in solving the path-finding problem above(Zhi Qiang)

Answer:

- Initial State : Node S
 - successor function : successor are all nodes that are connected to any node
[e.g. Node S's successor = { A , B , C }]
 - goal test : a function to check if current state is equal to Node G
 - step cost : the distance (km) from one node to another node that are both connected
[e.g. Node S and Node B are connected with a distance of 3 (in km)]
 - path cost : the sum of cost starting from initial State (Node S) to any node.
[e.g. Node S to Node E == travel from S - A - E == 4 + 4 = 8]

- Shortest Path = S → A → D → F → G
 Distance = 14 km

$f(n) = g(n) + h(n)$
 $g(n)$ = path cost
 $h(n)$ = Euclidean (Straight line) distance



c)

Time Efficiency: high
 Space Efficiency: high
 Optimality : optimal
 Completeness: yes

4. Consider 2 heuristic h_1 and h_2 of A* for the puzzle problem are defined as:
 $h_1(n)$ = number of misplaced tiles
 $h_2(n)$ = total Manhattan distance

3	1
2	

Start State

Goal State

1	2
3	

- a) Illustrate the **state space** of the puzzle to reach the goal state based on:
- ☐ $h_1(n)$ (Qi Xian)
 - ☐ $h_2(n)$ (Lik Sheng)
- b) Show the resulting search trees of A* search to find the shortest path using the heuristic functions of:
- ☐ $h_1(n)$ (Juo Sheng)
 - ☐ $h_2(n)$ (Divya)

(Remark: Avoid repeated state)

You must clearly show the function cost, given that: $f(n) = h(n) + g(n)$, where $g(n)$ is the path cost.

Answer:

a) i)

$$f(n) = h_1(n) + g(n)$$

3	1
2	

$$f(n) = 3 + 0 = 3$$

$$f(n) = 3 + 1 = 4$$

3	
2	1

3	1
	2

$$f(n) = 3 + 1 = 4$$

$$f(n) = 3 + 2 = 5$$

	3
2	1

	1
3	2

$$f(n) = 2 + 2 = 4$$

1	
3	2

$$f(n) = 1 + 3 = 4$$

1	2
3	

$$f(n) = 0 + 4 = 4$$

a) $h_2(n)$ (Lik Sheng)

$h_2(n)$ Manhattan Distance

$T_3: 0,0 \rightarrow 1,0$
 $(0-1) + (0-0) = 1$
 $T_1: 0,1 \rightarrow 0,0 = 1$
 $T_2: 1,0 \rightarrow 0,1$
 $(1-0) + (0-1) = 2$
 $g=1 \quad h=3 \quad f=4$

$g=0 \quad h=4 \quad f=0+4=4$
 $T_3: 0,0 \rightarrow 1,0 = 1$
 $T_2: 1,0 \rightarrow 0,1 = (1-0) + (0-1) = 2$
 $T_1: (1,0) \rightarrow (0,1) = (1-0) + (0-1) = 2$
 $g=1 \quad h=5 \quad f=6$

$T_3: \checkmark$
 $T_2: (1,1) \rightarrow (0,1) = 1$
 $T_1: (0,1) \rightarrow (0,0) = 1$
 $g=2 \quad h=2 \quad f=2+2=4$

$T_3: \checkmark$
 $T_2: (1,1) \rightarrow (0,1) = 2$
 $T_1: \checkmark$
 $g=3 \quad h=2 \quad f=3+2=5$

$T_3: \checkmark$
 $T_2: \checkmark$
 $T_1: \checkmark$
 $g=4 \quad h=0 \quad f=4+0=4$

$T_3: \checkmark$
 $T_2: (1,1) \rightarrow (0,1) = 2$
 $T_1: \checkmark$
 $g=3 \quad h=2 \quad f=3+2=5$

b) $h_1(n)$

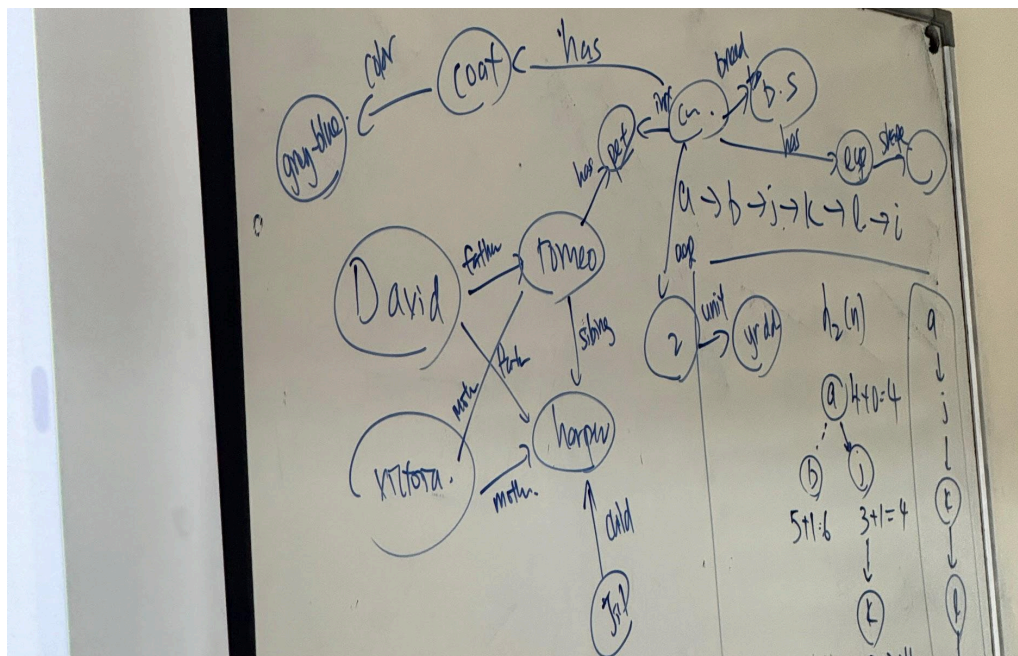
I am confused about difference/concept between state space and search tree.

Tutorial 5

1. Semantic network is a knowledge base that represents semantic relations between concepts in a network. Draw a semantic network for the following facts and relations. (Chee Luo)

"David and Victoria are the parents of Romeo and Harper. Romeo and Harper are brother and sister. Jane is the child of Romeo, and Jill is the child of Harper. Romeo has a pet called 'Christal,' a British Shorthair. Christal is 2 years old now and has a grey-blue coat, pineapple eyes, and a medium-sized tail."

Answer:



2. How are semantic networks different from Conceptual Graph? Explain your answer with the help of ONE (1) diagram for each of the methods based on the following statement: (Han Sheng)

"A cat is on a mat."

Answer:



	Semantic	Conceptual
Structure type	Informal graph	Formal logic-based graph
Representation Relation	Label on an arrow	Dedicate relation node ()
Formality	Less strict	Highly structured, rule-based

3. The following statements are given.

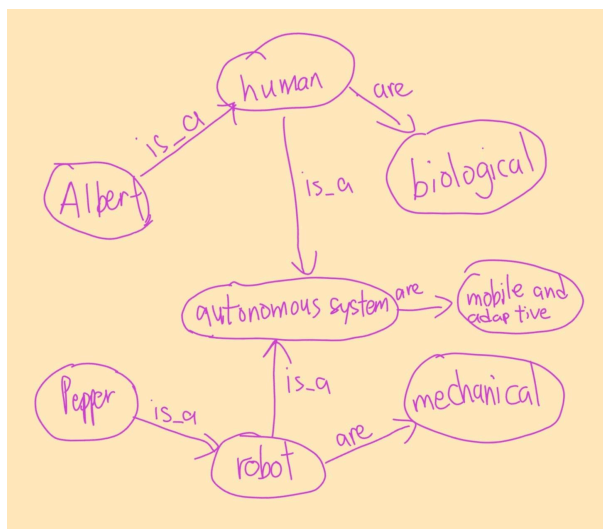
Albert is a human.
 Pepper is a robot.
 The construction of human is biological, while robot is mechanical.
 Both human and robot are autonomous system.
 The behaviours of autonomous system are mobile and adaptive.

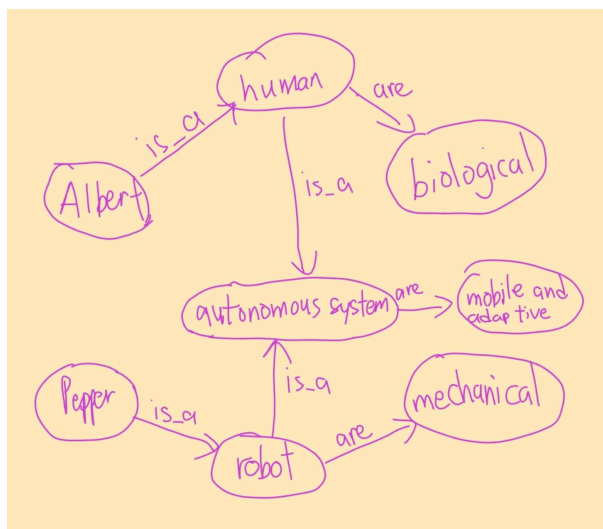
- a) Between semantic network and frames, justify which knowledge representation tool you would use to represent the information above. (Xuan Ping)
- b) Discuss **ONE (1)** limitation of the knowledge representation tool that is selected in Question 3(a) above. (Jin Zhi)
- c) Illustrate the representation of the given statements above using the knowledge representation tool that you selected in Question 3(a) above. (Zhen Wei)

Answer:

a) Frames - coz both Albert and Pepper are instances of autonomous system which allows them to inherit mobile and adaptive behavior from parent class to reduce redundancy.

b) One limitation of Frames is the difficulty in handling exceptions, as overriding default values for specific instances complicates the inheritance hierarchy and reasoning process.



c)  scam , this it not frame (TT—TT)

4. The following statements are given.

Bird is a living being that builds nest.
 Insect is an invertebrate living being.
 Ant is a kind of insects that builds nest.
 Both dragonfly and fly are insects that have wings.

- a) Between semantic network and frames, justify which knowledge representation tool you would use to represent the information above. (Divya) Discuss **ONE (1)** limitation of the knowledge representation tool that is selected in Question 4(a) above. (Yee Xian)

- b) Illustrate the representation of the given statements above using the knowledge representation tool that you selected in Question 4(a) above. (Shan Yi)

Answer:

(a) Semantic Network

↳ **Emphasis on relationships:** Focus on hierarchical relationships and properties that connect different concepts.
: Excel at representing such relationships through labelled links b/w nodes

↳ **Simple interconnected facts:** The information consist of straightforward facts about birds, insects and their subtypes (anti, dragonfly, fly) and their properties (build, nest, how wings, invertebrate).

b) limitation of a Semantic Network is that the relationship is not strictly defined, as a result, the same link may be interpreted differently by different users or systems, and ambiguity might happen when the network becomes large and complex.

5. Suggest and describe a real-life application where semantic network is applied. (Wei Sheng)

Answer:

A real-life application where a **semantic network** is effectively applied is in **Virtual Personal Assistants (VPAs)**, such as Siri, Google Assistant, and Alexa.

Virtual Personal Assistants use semantic networks as a core component of their knowledge representation and natural language understanding systems.

For example, when a user asks a question like, "**What is the population of the capital city of France?**". The VPA first uses Natural Language Processing (NLP) to identify the key concepts (nodes): "population," "capital city," and "France." It then traverses the semantic network using the relationships (edges) to find the answer:

-Find the node "France."

-Follow the "has_capital" relationship to the node "Paris."

-Follow the "has_population" relationship from "Paris" to the value (e.g., 2.14 million).

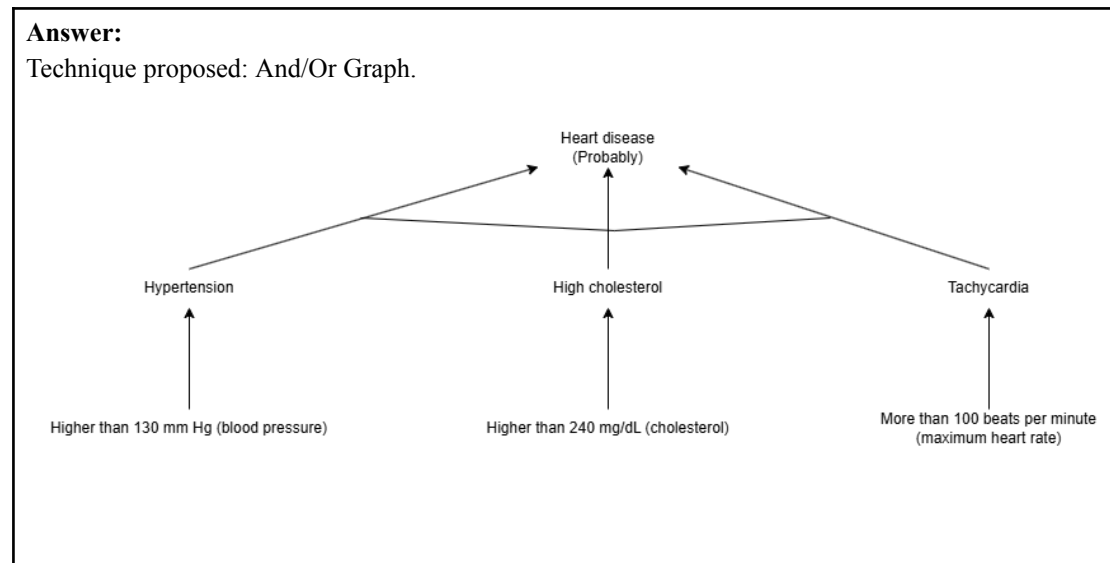
6. A rules based heart failure rule-based system is shown below.

- | | | |
|-----|------|--|
| R1: | IF | Blood pressure is higher than 130 mm Hg |
| | THEN | Most likely it is hypertension. |
| R2: | IF | Cholesterol is higher than 240 mg/dL |
| | THEN | Most likely it is high cholesterol. |
| R3: | IF | Maximum heart rate is more than 100 beats per minute |
| | THEN | It may be Tachycardia. |
| R4: | IF | There is hypertension |
| | AND | High cholesterol |
| | AND | Tachycardia |
| | THEN | Most probably the patient is having heart disease. |

Proposed **ONE (1)** technique to represent the knowledge above. Then, represent the knowledge by using the proposed technique. (Re Shun)

Answer:

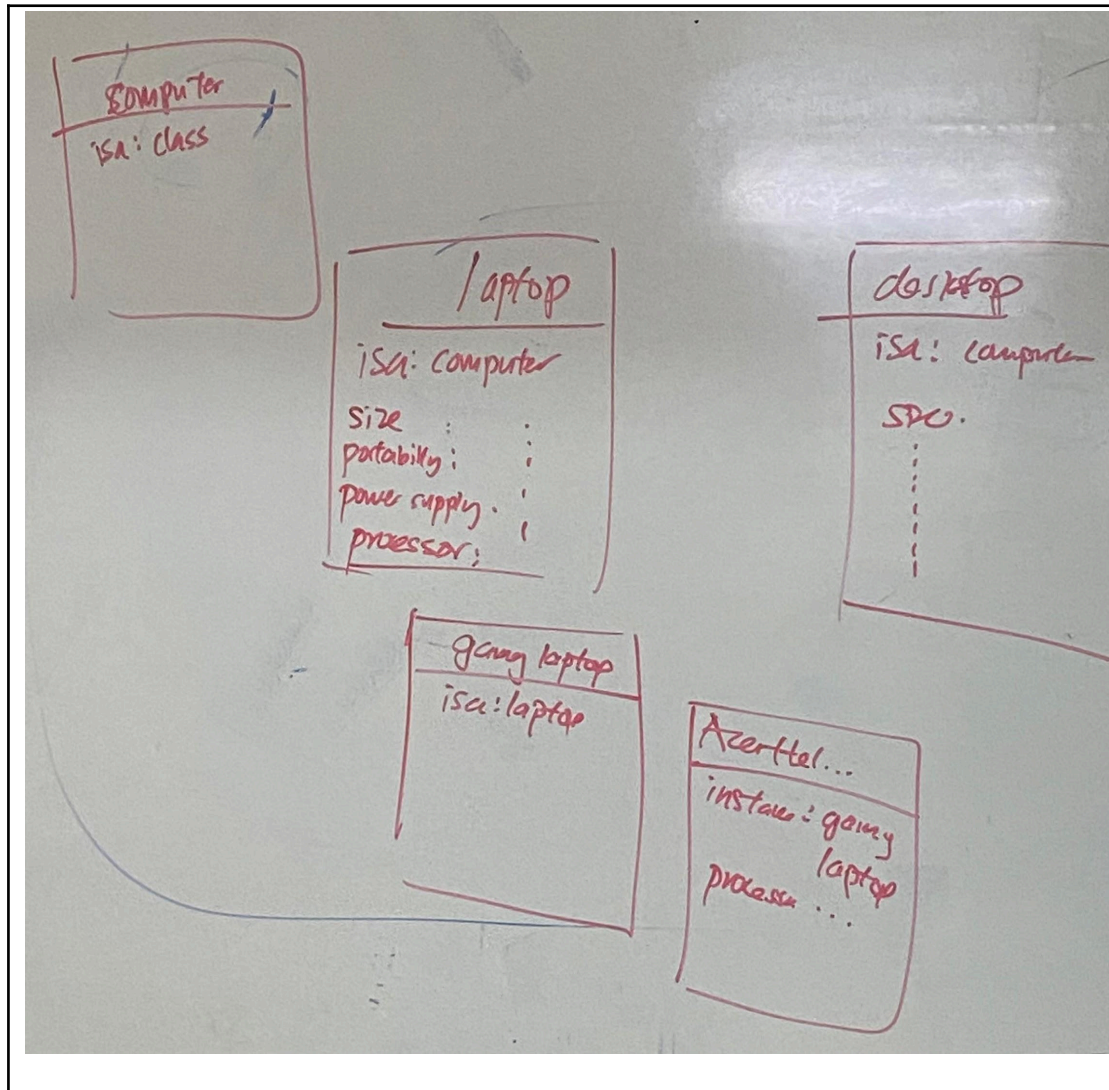
Technique proposed: And/Or Graph.



7. The information below describes the differences between a desktop and a laptop. Illustrate the information below with frames. (Jia Hang)

Laptop and desktop are the commonly used computer among students. Laptop is generally small and easily portable. It can run either on battery or main power supply. Desktop, on the other hand, is large and not portable. It only runs on main power supply. Besides that, desktop normally has more powerful processor as compared to laptop. ACER Predator Helios 500, which is a gaming laptop, however, is equipped with a powerful Intel i9 processor.

Answer:



Tutorial 6

1. Discuss the **THREE (3)** major obstacles involved in Natural Language Understanding (NLU). (Zhi Qiang)

Answer:

Ambiguity: A single word can have multiple meanings, "board" can be a noun or a verb, it depends on the sentence.

Heavily depends on Knowledge within the domain of the discourse:
she stole my heart ,we cannot chain the dictionary meanings of steal to the sentence

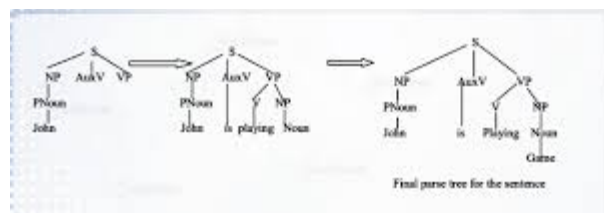
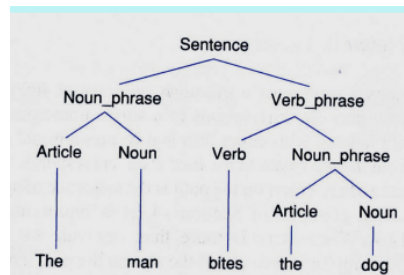
Machines difficult to understand implied meaning

2. What are the three typical stages involved in a Natural Language Processing (NLP) application development? Elaborate on these three stages with the aid of appropriate diagrams. (Yan Kang)

Answer:

1. Parsing

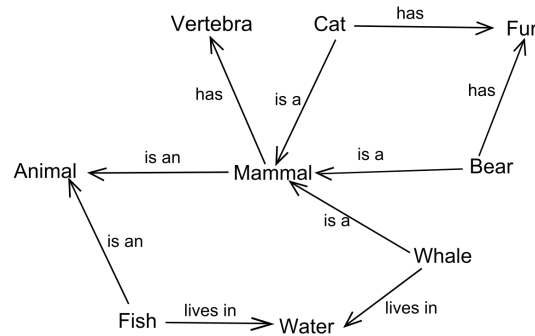
- Parsing analyses the syntactic structure of a sentence, and identifies the major relations of words such as subject-verb, verb-object and noun-modifier, then maps them together.



2. Semantic Interpretation

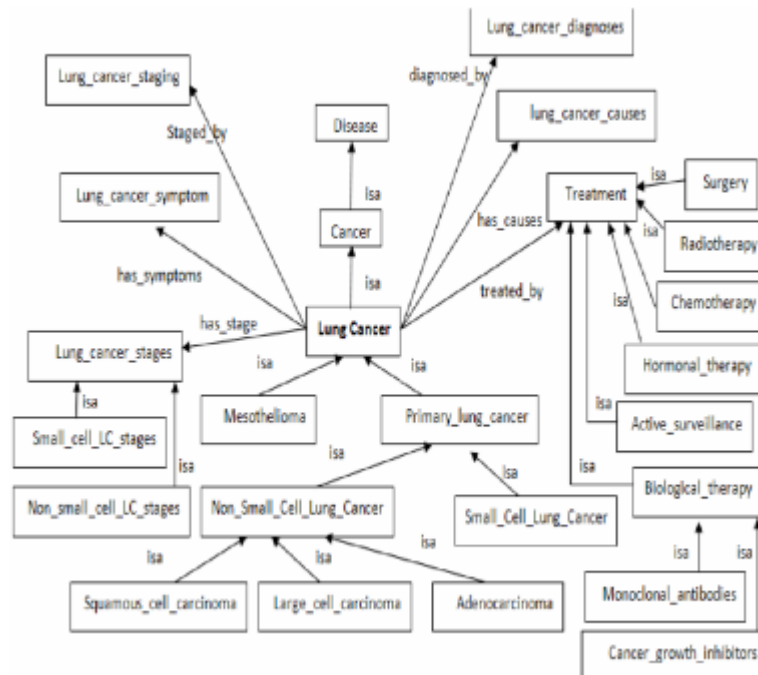
- Semantic Interpretation uses semantic network, conceptual graph, conceptual dependencies, frame, and more to understand the meaning and linguistics of words based on given knowledge.





3. World Knowledge Representation

- WKR uses methods and structures to model and store information about the real world that can be understood and utilised by NLP.



Albert Einstein was a German-born theoretical physicist who developed the theory of relativity.

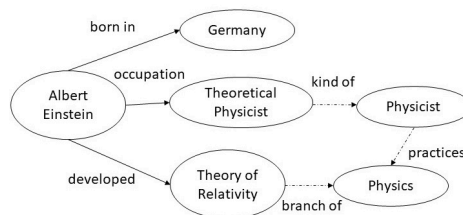


Figure 6: A knowledge graph created using entity and relation extraction

- Besides the three main phases involved in NLP as discussed in Question 2, describe three other significant analyses that can be done to improve NLP applications. (Qi Xian)

Answer:

i) Syntax

-rules for combining words into legal phrases and sentences

ii) Semantics

-the meaning of words, phrases, and sentences

iii) Morphology

-components that make up words, such as prefixes (un-, non-, anti-) and suffixes (-ing, -ly) that modifying the meaning of root words

4. NLP application requires the use of knowledge about human languages. Suggest an example of NLP application and describe the difficulties that a researcher would face during the development of the application that you have suggested. (Lik Sheng)

Answer:

Google Translate, its difficulty is ambiguity. Some words have multiple meanings

5. In NLP, representation is important as it can solve issues like the canonical form of sentences and the syntactic problems of a sentence.
- Explain the meaning of the canonical form of sentences. Provide examples to elaborate on your answer. (Juo Sheng)- **not important**
 - By referring to the statement below, there is a syntactic problem. Identify the problem and then draw **TWO (2)** different representations (by selecting either the semantic network or the conceptual graph) to solve the syntactic problem. (Chi Ban)

"Rambo hit the man with a hammer"

Answer:

- a) Two different sentences with the same meaning.

Example:

I was given this watch by my dad.

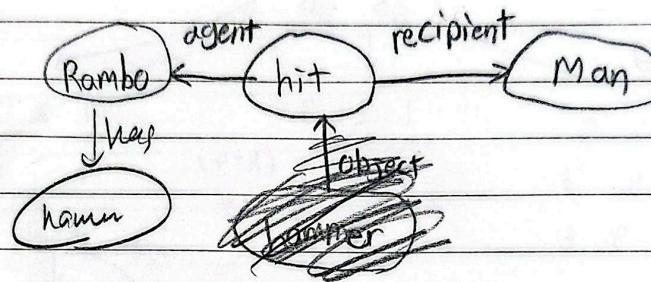
My dad gave me this watch.

B) This sentence has a prepositional phrase (PP) attachment ambiguity. The PP "with a hammer" can attach to the verb phrase (instrument - Rambo used a hammer) or to the noun phrase (modifier - the man had the hammer). These are two distinct meanings.

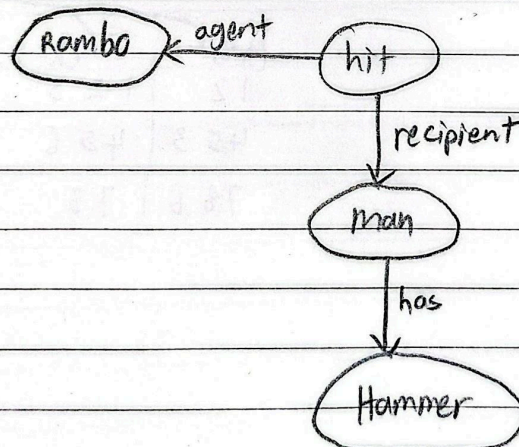
Date.

Semantic Representation (1)

Rambo used a hammer to hit the man.



2) Rambo hit the man who had a hammer.



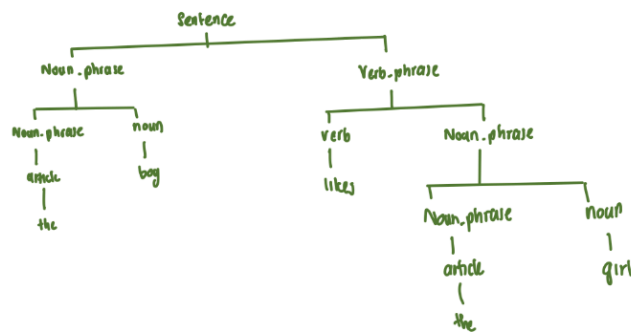
6. Parse tree is a popular tool used in one of the phases of NLP called parsing. (Divya)
- Explain the importance of parsing in NLP.
 - With the aid of the simple English grammar for simple transitive sentences as shown next page, draw the respective **parse trees** to verify the sentences "**the boy likes the girl**" and "**time flies like an arrow**"

Sentence	-> noun_phrase verb_phrase
Noun phrase	-> noun
Noun-phrase	-> article noun
Verb phrase	-> verb noun_phrase
Verb phrase	-> verb preposition noun_phrase
preposition	[like]
article	[a, an, the]
noun	[flies, time, arrow, boy, girl]
verb	[like, flies]

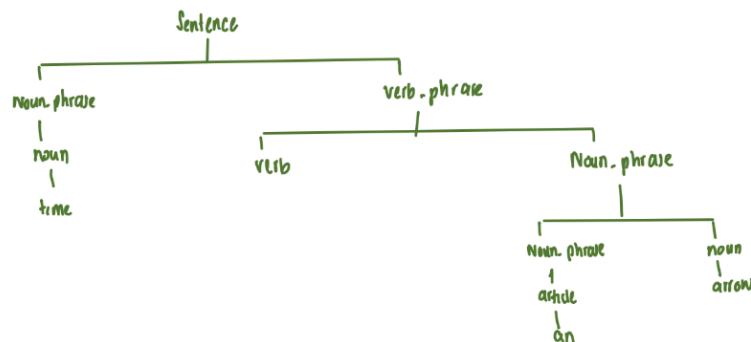
Answer:

(b) Parse tree

1. the boy likes the girl



2. time flies like an arrow



(a) Understanding Sentence Structure

- Parsing analyzes the grammatical structure of sentences, breaking them down into their constituent parts.
 - Help computer understand how words relate to each other within a sentence

↓ Disambiguation:
 Natural language is often ambiguous. Parsing helps resolve structural ambiguities by determining the correct grammatical relationship

7. Considering the sentence S = "She beats George with one hand at the bank".
- The sentence S consists of semantic ambiguity and syntactic ambiguity. Identify both of the ambiguities found in the sentence above. (Yee Xian)
 - Given the grammar below, construct **ONE (1)** parse tree for the sentence S. (Remark: grammar in the parentheses () means it is optional.)(Shanyi)

S -> NP VP (PP)
NP -> (DET) (ADJ) N (PP)
VP -> V NP (PP)
PP -> P NP
ADJ -> [one]
DET -> [a, the]

N -> [She, George, hand, bank]
V -> [beats]
P -> [with, at]

Answer:

7a) Semantic Ambiguities:

- i) The word “bank” have can mean they are at a **financial institution or a riverbank**
- ii) The word “beats” can mean she **defeats** or **physically hit George**
- iii) The word “hand” can mean **a round of card game** or the **hand** itself literally

Syntactic Ambiguities:

with one hand and **at the bank** can attach to different parts of the sentence, leading to different meanings.

- i) (She beats George) (with one hand) (at the bank)

- She use only one hand to beat George

- ii) (She beats) (George with one hand) (at the bank)

- George has one hand, and she beats George

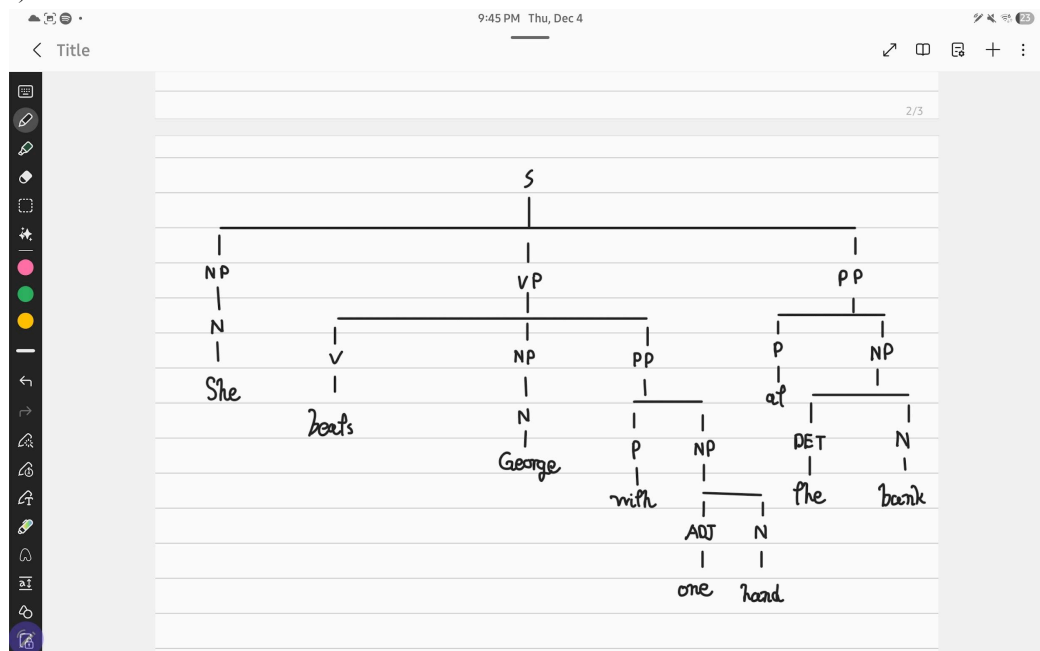
- iii) (She beats George with one hand) (at the bank)

- The action occurs either at the financial institution or river bank

- iv) (She beats) (George with one hand at the bank)

- She beats George where George is at the bank.

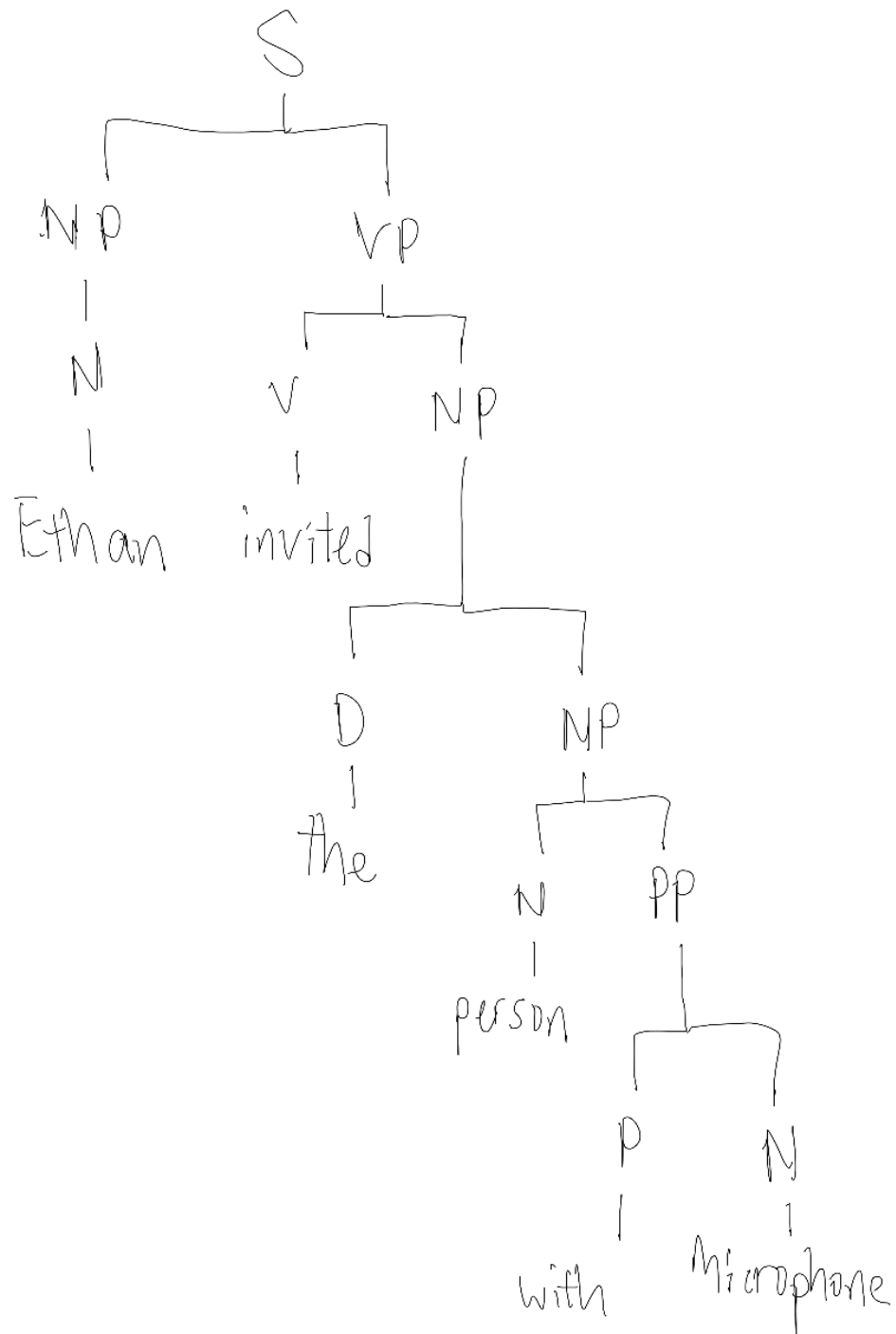
b)



8. Consider a sentence, $S = \textit{Ethan invited the person with microphone}$. (xuanping)
- The sentence S consists of syntactic ambiguity. Identify the ambiguity found from the sentence above.
 - Given the grammar below, construct **ONE (1)** parse tree for the sentence S .

```
noun_phrase (NP)
verb_phrase (VP)
preposition_phrase (PP)
determiner (D)
noun (N)
verb (V)
S -> NP VP
NP -> N | N PP | D NP | D N
VP -> VP PP | V NP
PP -> P N
N -> [Ethan, person, microphone]
V -> [invited]
P -> [with]
D -> [the]
```

Answer:



Tutorial 7

1. Differentiate between artificial intelligence (A.I.) and machine learning (ML). (Chi Ban)

Answer:

Artificial Intelligence (AI) is the field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.

These tasks include understanding language, recognizing images, reasoning, learning, solving problems, and making decisions.

Machine Learning is a set of methods that can automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other kinds of decision making under uncertainty

2. There are **TWO (2)** main types of ML, which are Supervised and Unsupervised Learning. (Yan Kang)
- Discuss the differences between them.
 - Provide **TWO (2)** algorithms for each type of ML.

Answer:

a)

Supervised ML	Unsupervised ML
Learn with data labelled with pre-defined classes	Learn with unlabelled data
Predict the dependent variable based on the independent variable	Find patterns among data in dataset

b)

Supervised ML	Unsupervised ML
Logistic Regression	K-Means Clustering
Support Vector Machine (SVM)	Hierarchical Clustering

3. Assume that the Sea Fisheries Department has approached you to analyse abalone from physical measurements. A set of data that contains 7 attributes and 4177 rows of records is given. Part of the data is shown in **Table 1** below:

Table 1: Partial Abalone Data*

Sex	Length (mm)	Diameter (mm)	Height (mm)	Whole weight (g)	Shell weight (g)	No. of rings
F	0.59	0.455	0.145	1.063	0.25	8
F	0.595	0.47	0.155	1.121	0.155	11
F	0.595	0.45	0.15	1.114	0.25	11
F	0.595	0.46	0.14	1.0045	0.2515	9
F	0.605	0.49	0.15	1.1345	0.295	9
M	0.595	0.475	0.165	1.213	0.274	9
M	0.595	0.455	0.15	1.044	0.27	9
M	0.605	0.475	0.155	1.161	0.275	9
M	0.605	0.47	0.165	1.2315	0.2925	11
M	0.61	0.47	0.15	1.1625	0.3085	11

Data Source: <http://mlr.cs.umass.edu/ml/datasets/Abalone>

Reference: Asuncion, A. & Newman, D.J. (2007). UCI Machine Learning Repository

[<http://www.ics.uci.edu/~mllearn/MLRepository.html>]. Irvine, CA: University of California, School of Information and Computer Science.

Between supervised learning or unsupervised learning, which type of ML is suitable to perform the following data analytics? Justify your answer. Then suggest **ONE (1)** suitable ML algorithm for each case.

- To determine 3 different grades of abalone based on diameter and whole weight. (Shan Yi)
- To predict the age of an abalone. (To determine the age of an abalone, we can check the number of rings, which is between 1 to 29) (Chee Luo)

Answer:

a) Unsupervised Learning

Justification

- Because there is no predefined label, the dataset doesn't provide grades, needed to discover from the data itself.
- clustering problem as we have to partition the abalone into 3 groups based on the physical measurement.

Suitable ML Algorithm: K mean clustering

- partition data into K distinct clusters
- group abalone together with similar physical measurement.
- efficient for grading and categorization

b) Supervised Learning

Justification:

- The dataset already contains the number of rings, which is the target output.
- Since age is determined by rings, this is a prediction problem with known labels.
- This makes it a regression task because the target output (number of rings) is a numeric value.

Suitable ML Algorithm: Linear Regression

- Simple and effective for predicting continuous numeric outcomes such as the number of rings.
- Commonly used as a baseline model for regression tasks

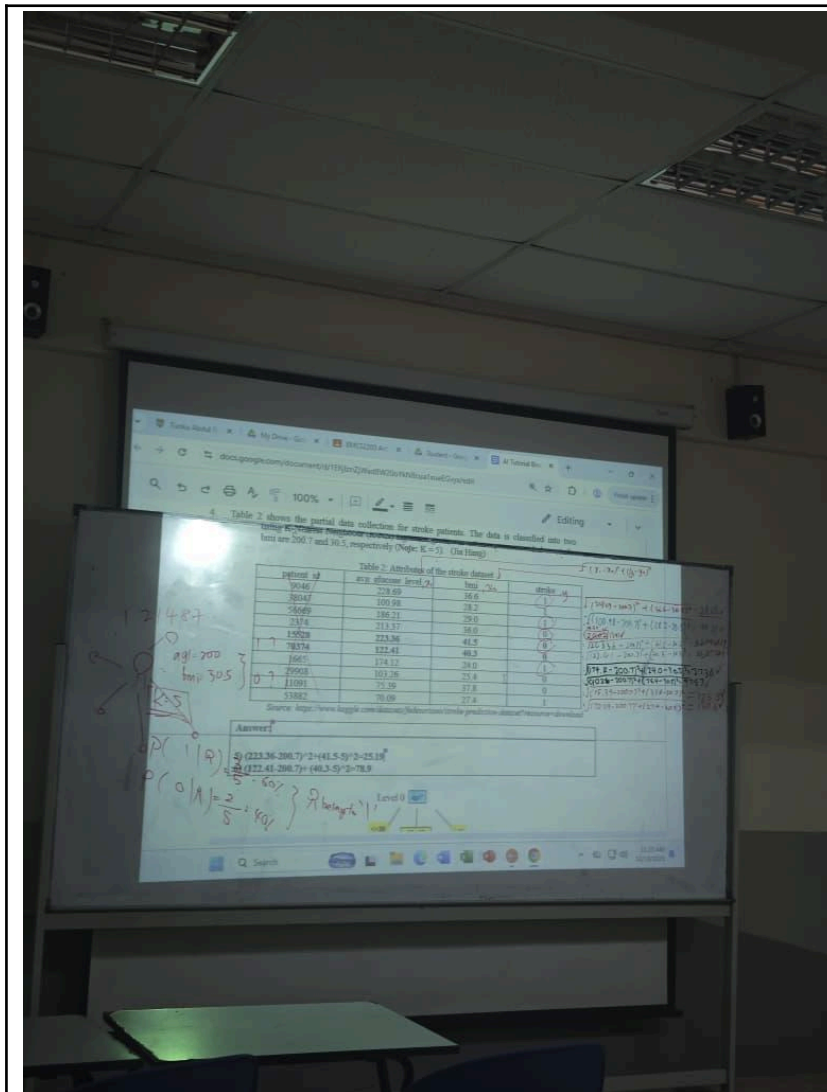
4. Table 2 shows the partial data collection for stroke patients. The data is classified into two different classes which are Yes (1), and No (0). Identify the class that a new patient belongs to by using K-Nearest Neighbour (K-NN) algorithm given that the new patient's avg_glucose_level and bmi are 200.7 and 30.5, respectively (**Note:** K = 5). (Jia Hang)

Table 2: Attributes of the stroke dataset

patient_id	avg_glucose_level	bmi	stroke
9046	228.69	36.6	1
38047	100.98	28.2	1
56669	186.21	29.0	1
2374	213.37	36.0	0
15528	223.36	41.5	0
70374	122.41	40.3	0
1665	174.12	24.0	1
29908	103.26	25.4	0
11091	75.39	37.8	0
53882	70.09	27.4	1

Source: <https://www.kaggle.com/fedesoriano/stroke-prediction-dataset?resource=download>

Answer:



- 14.57
- 13.81
- $(223.36 - 200.7)^2 + (41.5 - 5)^2 = 25.19$
- $(122.41 - 200.7)^2 + (40.3 - 5)^2 = 78.9$

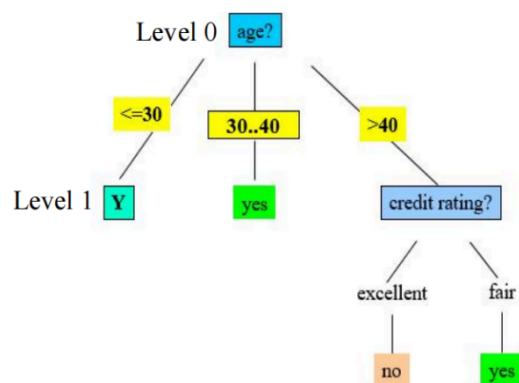


Figure 3: Partial decision tree of buys_computer

Table 2: The decision table of buys_computer

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

5. Given the table and the corresponding decision tree as above.
- Determine the appropriate attribute Y (income, student or credit_rating) at level 1 (Figure 3) by using Information Gain. Justify your answer. (Wei Sheng)
 - Produce the complete decision tree based on the answer from Question 6(a). (Xuan Ping)
 - Given a new sample, $X = \langle \leq 30, \text{low}, \text{no}, \text{excellent} \rangle$, predict the class it belongs to. (Juo Sheng)
 - Refer to **Table 2**, given a new sample, $X = \langle \leq 30, \text{low}, \text{no}, \text{excellent} \rangle$, predict the class it belongs to, $P(C|X)$, using Naïve Bayes classification. (Jin Zhi)

To calculate the information gain:

$$I(p, n) = -\frac{p}{p+n} \log_2 \frac{p}{p+n} - \frac{n}{p+n} \log_2 \frac{n}{p+n}$$

$$E(A) = \sum_{i=1}^v \frac{p_i + n_i}{p+n} I(p_i, n_i)$$

$$\text{Gain}(A) = I(p, n) - E(A)$$

Answer:

c) No because people that is aged 30 and below that is not student usually does not buy computer based on the data given.

Figure 3: Partial de

Since $P(\text{no}|D) > P(\text{yes}|D)$
 $\therefore X$ will not buy PC

compare $\left\{ \begin{array}{l} P(K_1|D) = \frac{P(D|h_1) \times P(h_1)}{P(D)} \\ P(K_2|D) = \frac{P(D|h_2) \times P(h_2)}{P(D)} \end{array} \right.$

$P(\text{no}|\leq 30, \text{low}, \text{no}, \text{ex})$
 $= P(\leq 30|\text{no}) \times P(\text{low}|\text{no}) \times P(\text{no}|\text{no}) \times P(\text{ex}|\text{no}) \times P(\text{no})$
 $= \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{2}{14}$
 $= 0.02$

$P(\text{yes}|D) = \frac{P(\leq 30|\text{yes}) \times P(\text{low}|\text{yes}) \times P(\text{no}|\text{yes}) \times P(\text{ex}|\text{yes}) \times P(\text{yes})}{P(D)}$
 $= \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14}$
 $= 0.005$

5. Given the table and the corresponding decision tree

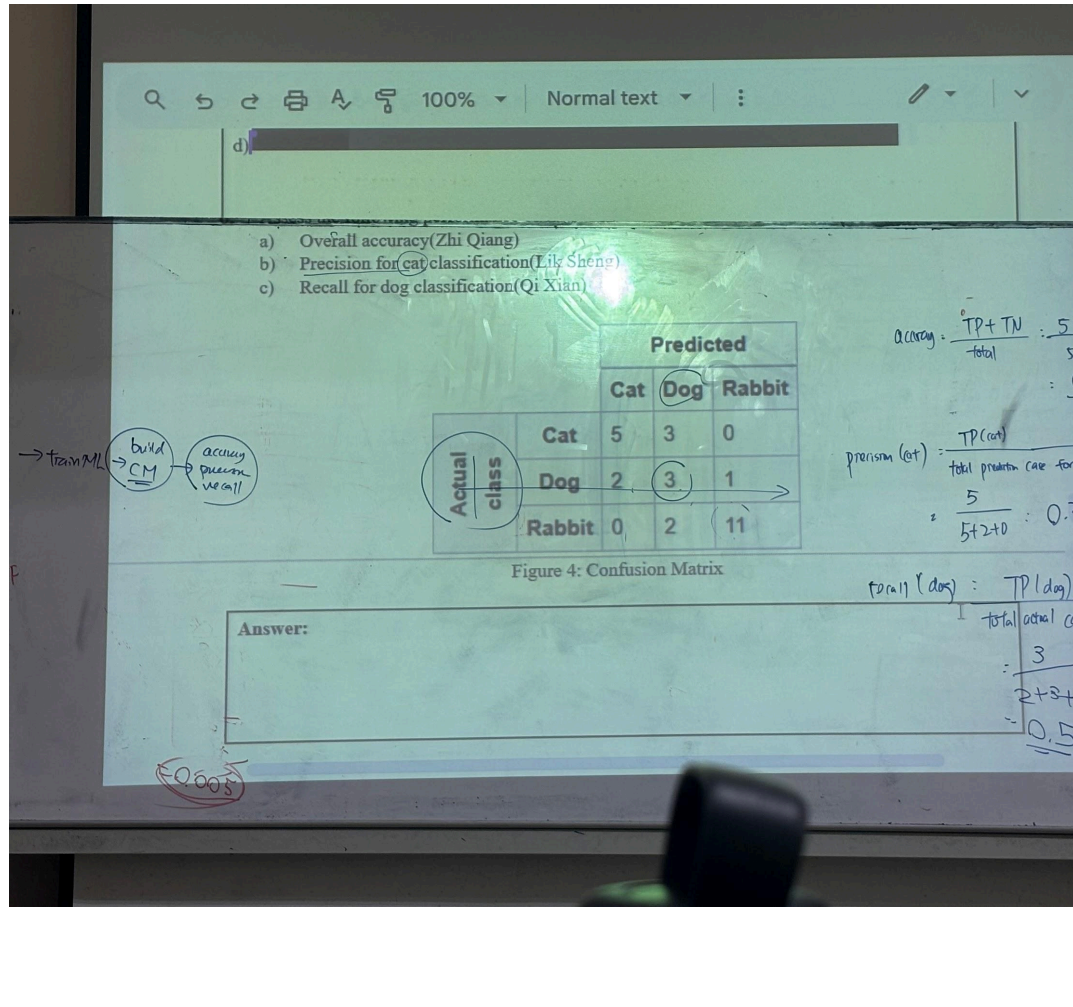
- Determine the appropriate attribute by using Information Gain. Justify your answer.
- Produce the complete decision tree based on the attribute chosen in (a).
- Given a new sample, $X = \langle \leq 30, \text{low}, \text{no}, \text{ex} \rangle$, what is the predicted class (Sheng)?
- Refer to Table 2, given a new sample belongs to, $P(C|X)$, using Naïve Bayes.

6. Assess the following performance of the classifier as shown in Figure 4:
- Overall accuracy(Zhi Qiang)
 - Precision for cat classification(Lik Sheng)
 - Recall for dog classification(Qi Xian)

		Predicted		
		Cat	Dog	Rabbit
Actual class	Cat	5	3	0
	Dog	2	3	1
	Rabbit	0	2	11

Figure 4: Confusion Matrix

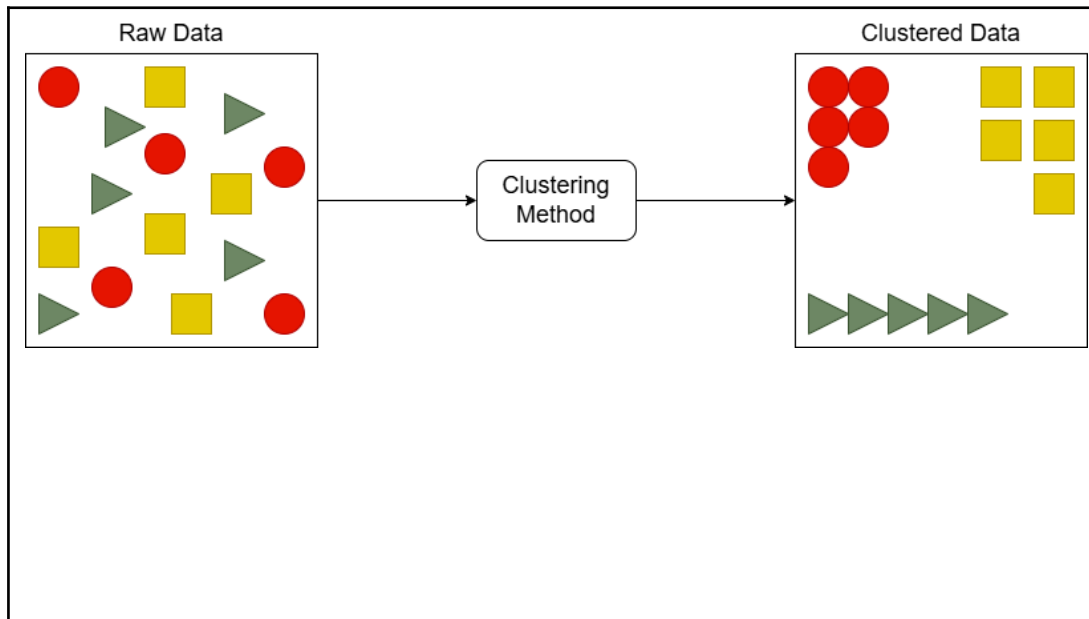
Answer:



Tutorial 8

1. Draw a diagram to depict unsupervised learning model. (Re Shun)

Answer:



2. Considering a partial dataset as shown in Table 1: (Yee Xian)
- Explain how regression model is different from clustering.
 - Explain how regression model and clustering can be applied on the dataset below.

Table 1: Sample of data

Age	Education	Gender	Income
31	Doctoral	Male	150k
45	Bachelor's	Female	78k
26	Master's	Male	76k
...	...	...	...

Answer:

a) Explain how regression model is different from clustering.

- **Regression model** is a supervised learning technique. It is used when there is a target variable to predict, based on one or more independent variables. The main goal is to model the relationship between variables and predict numeric values.
- **Clustering** is an unsupervised learning technique. It does not use a target variable. The goal is to group data points into clusters such that individuals within the same cluster are similar to each other, and different from those in other clusters.

b) Explain how regression model and clustering can be applied on the dataset below.

Regression Model

- Target variable: Income
- Independent variables: Age, Education, Gender
- A regression model can be used to predict a person's income based on their age, education level, and gender.
- Example use: Estimating expected income for a 30-year-old with a Master's degree.

Clustering

- No target variable is required.
- The dataset can be clustered based on Age, Education, Gender, and Income.
- This can group individuals into segments such as:
 - Young professionals with medium income
 - Highly educated, high-income individuals
 - Older individuals with moderate income
- Example use: Identifying customer or population segments for analysis or policy planning.

3. Clustering is the grouping of unlabeled data, it can be used for knowledge discovery. Discuss about this and provide **TWO (2)** real world applications of clustering. (Han Sheng)

Answer:

4. K-means is used to cluster the data in Table 2. Randomly we choose two cluster centers ($k = 2$) for two clusters; $C1 = (1.0, 1.0)$ and $C2 = (5.0, 7.0)$.

Table 2: Sample of data

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

- a) List the clusters that each data will be assigned to. (Zhen Wei)
- b) Identify the new centers for the first iteration. (Divya)
- c) Suggest **TWO (2)** stopping conditions for K-means. (Yan Kang)
 1. no (or minimum) re-assignment of data points to different clusters
 2. no (or minimum) changes of centroids
- d) Discuss on the limitation of K-means. (Chee Luo)
 - Very sensitive to the initial points: Do many runs of k-Means, each with different initial centroids.
 - Must manually choose k: Learn the optimal k for the clustering
 - K-means has problems when clusters are of differing sizes, densities and non-globular shapes
 - K-means has problems when the data contains outliers.

5. Mean Shift is one of the popular clustering algorithms: (Chi Ban)
- Demonstrate the algorithm for Mean Shift
 - Discuss the advantages and disadvantages of Mean Shift.

Answer:

6. Explain why cluster evaluation is a hard problem. Identify **TWO (2)** methods based on internal information that we can use to evaluate a clustering algorithm. (Jia Hang)

Answer:

Tutorial 9

1. Answer the following questions:
- Research in Artificial Neural Networks (ANNs) started with a view of mimicking the functioning of Biological Neural Networks (BNNs). Illustrate with diagrams, how the BNNs and ANNs are similar to each other. ()
 - Figure 1 below shows the simplest neural network model that is known as perceptron, which can be used for classification tasks. Briefly explain the functions of each of the layers A, B and C as shown in the diagram. ()

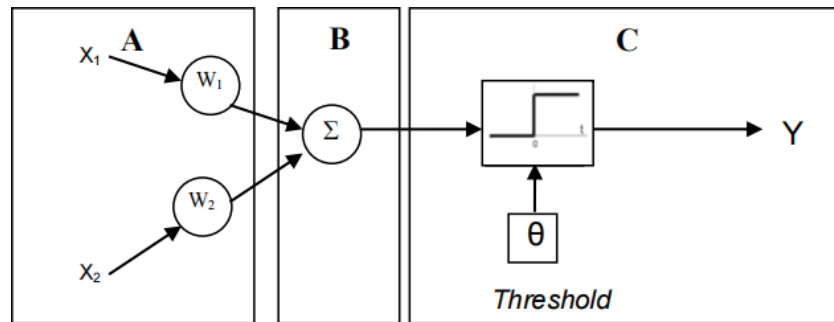


Figure 1: A sample neural network model

- c) Back propagation Multilayer Artificial Neural Network can be applied into a system that is able to recognize and interpret two different types of flowers. Discuss the prerequisite for the Neural Network system and design **ONE (1)** Neural Network structure for the purpose. State the advantage of back propagation network. ()

Answer:

2. Assume that the Sea Fisheries Department has approached you to analyze abalone from physical measurements. A set of data that contains 7 attributes and 4177 rows of records is given. Part of the data are shown in Table 1 below:

Table 1: Partial Abalone Data

Sex	Length (mm)	Diameter (mm)	Height (mm)	Whole weight (g)	Shell weight (g)	No. of rings
F	0.59	0.455	0.145	1.063	0.25	8
F	0.595	0.47	0.155	1.121	0.155	11
F	0.595	0.45	0.15	1.114	0.25	11

Demonstrate how a **multi-layer feedforward back-propagation neural network** can be constructed to predict the number of rings of abalone. Illustrate the architecture of the neural network with appropriate labels of each layer in the diagram. In your diagram, also include all the attributes and output as shown in Table 1. ()

Answer:

3. A combinational logic circuit are made up from basic logic AND, OR, NAND, NOR or NOT gates that are “combined” or connected together to produce more complicated switching circuit. The output of the gate will be determined by combination of 1's and 0's which are present at the input terminals. The truth table of 3-input combinational logic circuit is given by Table 2 and Figure 2 where A, B and C are the inputs and Q is the output. ()

Table 2

A	B	C	Q
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

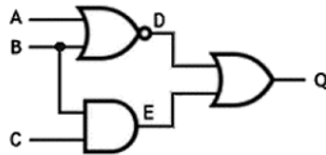


Figure 2

Train the perceptron to perform 3-input combinational logic circuit operation. The formula of output Q is given by

$$Q = w_1A + w_2B + w_3C$$

where w_1 , w_2 and w_3 are weight for A, B and C respectively. Assume that the threshold $\theta = 0.25$, learning rate $\alpha = 0.1$ and **step function** is used for activation function. Calculate the final weights of w_1 , w_2 and w_3 for epoch 1 only. (Assume the initial weights $w_1 = 0.3$, $w_2 = 0.2$ and $w_3 = 0.1$)

Answer:

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4. In a supervised learning session, a 2-input neuron and the respective truth table are given in Figure 3 and Table 3, where x_1 and x_2 are the inputs and y is the output. ()

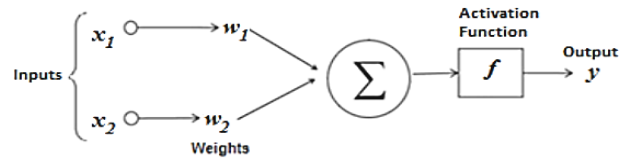


Figure 3: The 2-input neuron

Table 3

x_1	x_2	Desired Output, y_d
0	0	-1
0	1	1
1	0	1
1	1	1

The formula of output y is given by

$$y = \text{sign}(n) = \begin{cases} 1 & \text{if } n \geq \theta \\ -1 & \text{if } n < \theta \end{cases}$$

where $n = w_1x_1 + w_2x_2$, and w_1 and w_2 are the weights for x_1 and x_2 respectively. The **sign function** is used for activation function. The weights, $w_1 = 0.1$ and $w_2 = 0.2$, the threshold, $\theta = 0.2$ and the learning rate, $\alpha = 0.1$ are set initially for the training. Train the neuron and demonstrate the final weights of w_1 and w_2 **for epoch 1 only**.

Answer:

Tutorial 10

1. Noise reduction is very important in image processing. ()
 - a) Discuss the above statement.
 - b) Identify **TWO (2)** types of noises in an image.
 - c) Name a filter to perform noise reduction.
 - d) Demonstrate how image filtering works in reducing the noise by using median filter. Perform median filtering onto the image in Figure below by using 3×3 kernel. (Note: Replace the pixel value to 0 for border problem due to not enough neighbouring pixels to perform the filtering).

11 5	14 5	14 5	10 9	13 5	14 6
16 3	17 8	20 0	10 9	15 7	13 8
16 3	19 8	20 1	11 8	14 4	14 5
16 3	19 8	20 1	10 9	13 5	12 4
15 7	19 8	20 1	11 8	14 4	10 9
14 3	15 3	17 1	17 3	17 0	16 8
13 7	14 4	16 8	14 3	16 1	14 1

Answer:

2. Morphological operation is important in image processing. ()
 - a) Explain **FOUR (4)** morphological operations.
 - b) Suggest a morphological operation that could be used to recover the image as shown in figure below (the objective is to close the gap between the pixels of the number). Explain your answer.



Answer:

3. Explain **ONE (1)** importance of morphological image analysis in image processing. ()

Answer:

4. In morphological image analysis, morphological operators often probe an image with a **structuring** element. A 6x6 image is given in **Figure 1** below. (Note: Replace the pixel value to 0 for border problem due to not enough neighbouring pixels to perform the filtering)

	0	1	2	3	4	5
0	0	0	0	0	0	1
1	0	1	1	1	1	1
2	0	1	1	1	1	0
3	0	1	1	1	1	0
4	0	1	1	1	0	0
5	1	1	1	0	0	0

Figure 1: 6x6 binary image

- a) Based on the 3x3 structuring element with the origin in the middle as shown in Figure 2, perform an erosion effect on the 6x6 binary image as shown in Figure 1. Provide the resulting binary image. ()

1	1	1
1	1	1
1	1	1

Figure 2: 3x3 square structuring element

- b) Based on the 2x2 structuring element with the origin in the middle as shown in Figure 3, perform an dilation effect on the 6x6 binary image as shown in Figure 1. Provide the resulting binary image. ()

0	1
1	0

Figure 3: 2x2 square structuring element

Answer:

5. The scanned document in Figure 4 shows a missing portion of pixels (with an actual value of 2) resulting from a poor scanning process. Figure 5 represents the pixel value of Figure 4. Utilising the structuring element depicted in Figure 5, apply both dilation and erosion operations to Figure 5. Then, develop a resulting complete pixel for each operation. The origin of the structuring element is denoted by the bold box in Figure 6. Also, conclude which technique is more suitable to restore the missing part of the Figure 4. (Note: Replace the pixel value to 0 for border problem due to not enough neighbouring pixels to perform the filtering) (Assign 2 students)



Figure 4: Scanned image

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0
0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	1	1	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0	0
0	0	1	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

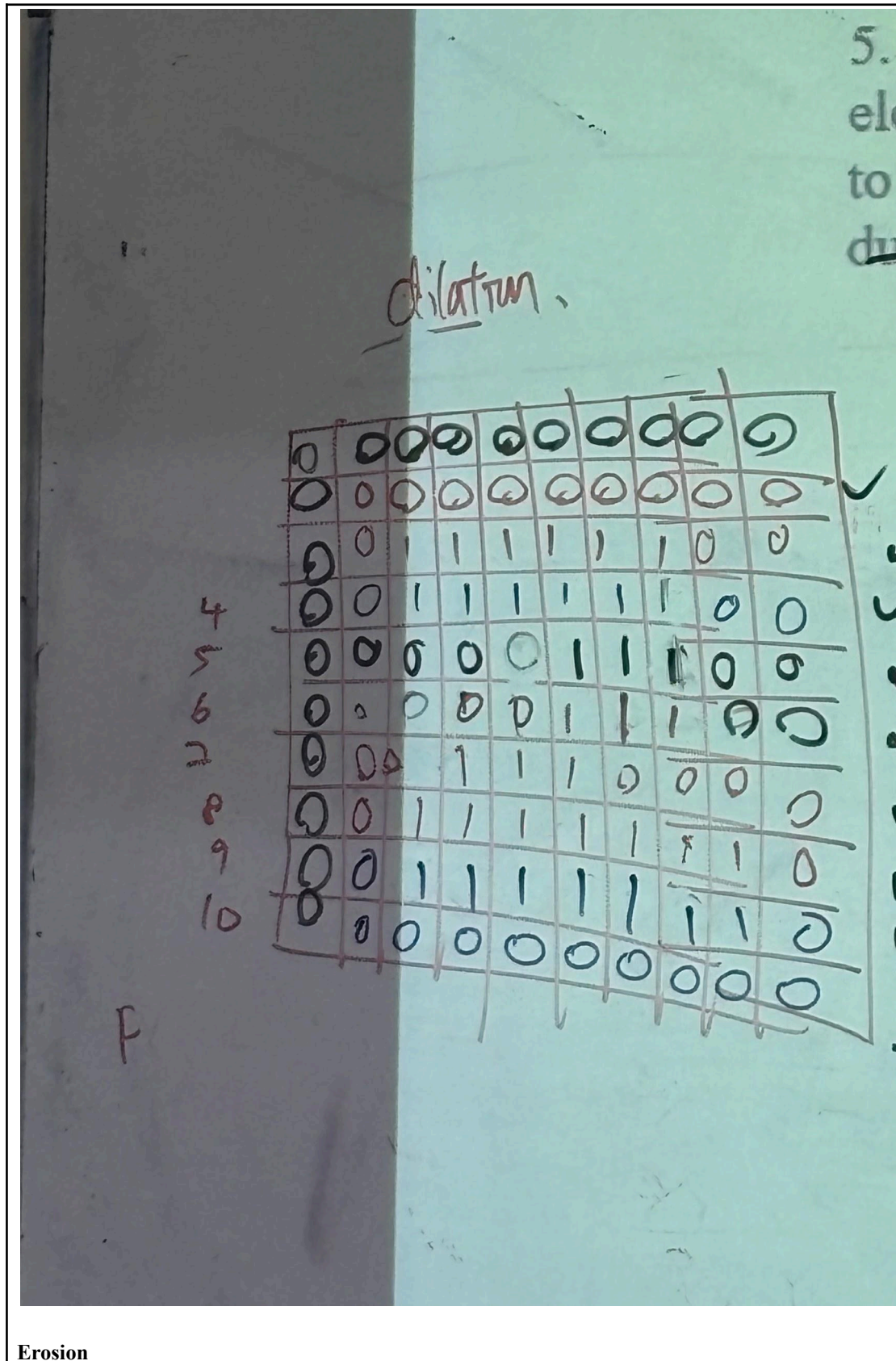
Figure 5: Pixel value

1	1
1	1

Figure 6: Structuring element

Answer:

Dilation



tion of pixels (with an actual value of 2)

intation and evolution of Figure 4. Utilising

operation. The origin of the structuring

conclude which technique is more suitable

the pixel value to 0 for border problem

ering) (Assign 2 students)

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
1	1	1	1	1	0	0	0
0	0	0	0	1	0	0	0
0	0	0	1	1	0	0	0
0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0
1	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

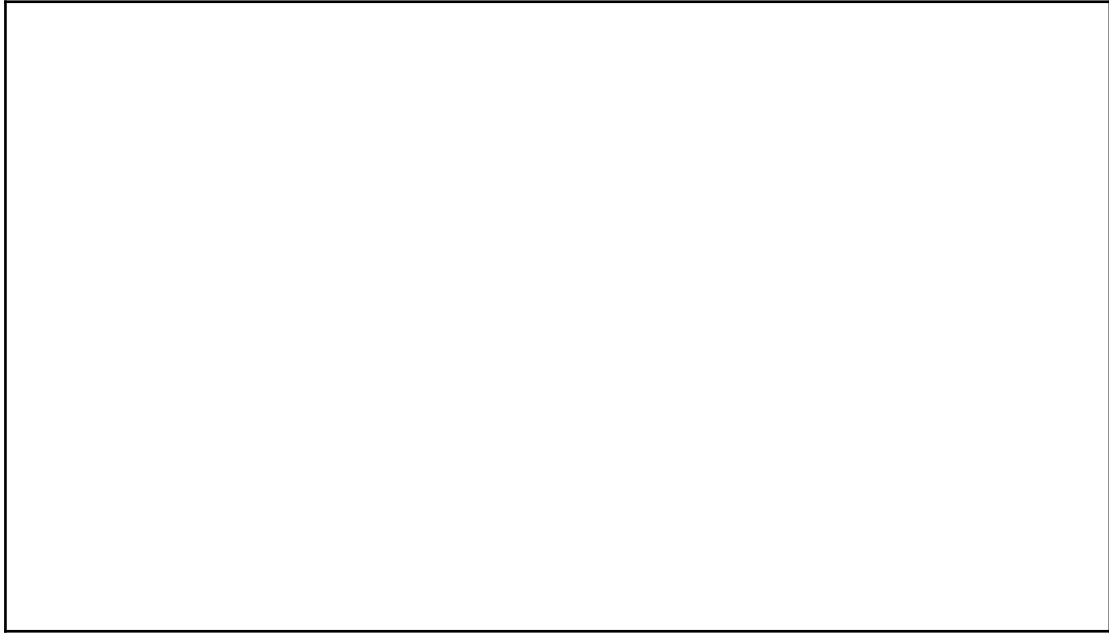
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Figure 5: Pixel value

ment

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6. Pre-processing is one of the important steps in digital image analysis. ()
- Explain the importance of image pre-processing.
 - Identify the suitable pre-processing operations based on the images in **Figure 7**. Provide brief description for each of the suggested operations.

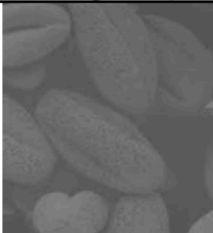
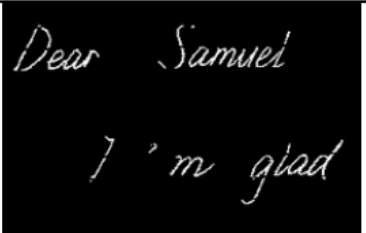
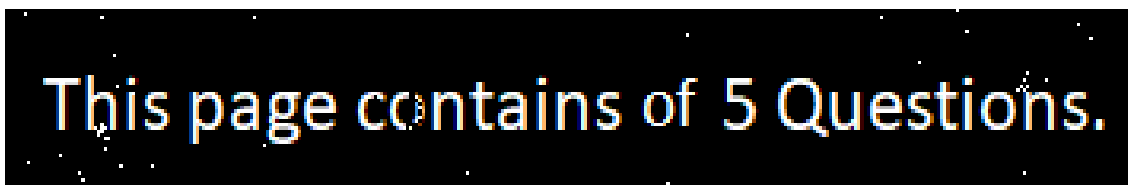
Image A (objects are less distinctive from its background)	Image B (salt and pepper in the image)	Image C (broken alphabet)
		

Figure 7: Sample images

Answer:

7. Figure below shows a part of scanned image which will be recognized by optical character recognition (OCR) algorithm. Nevertheless, the detection performance is relatively poor compared to the actual sentence “This page contains of 5 Questions.” Identify TWO (2) potential output error produced by OCR algorithm. In an addition, provide ONE (1) preprocessing technique to increase the accuracy of OCR performance for each potential output error. ()



Answer:

Tutorial 11

Part I: Expert Systems

1. Given a customer has problem deciding whether or not to purchase a car. He consulted an expert for advice and the consultant suggested some rules as follows.

Rule 1: IF The condition of car is poor
 OR The price of the car is high
 THEN Don't buy the car

Rule 2: IF Mileage on the car exceeds 100,000
 AND The car is city driven
 AND The body of the car is bad
 THEN The condition of the car is poor

Rule 3: IF The car has dents
 THEN The body of the car is bad

Rule 4: IF The car has rust
 THEN The body of the car is bad

- a) Assume the hypothesis is "Don't buy the car", draw an inference network (and-or graph) that supports the problem above. ()
b) Given the observation by the expert as follows, demonstrate the inference process by using forward-chaining. ()

Mileage: 160,000 miles Location of car: KL city Dents on car: right rear door and back door Rust on car: not noticed Price of car: \$1000 (remark: considered low)
--

- c) If you are going to develop the above expert system, which inference mechanism are you going to implement? Forward or backward chaining? Justify your answer. ()

Answer:

2. A basic structure of a rule-based expert system involves the components such as knowledge base, database, inference engine, explanation facilities, user interface, and user. ()
- Draw a diagram to illustrate the relations between the components in an expert system as given above, and describe the functions of each component.
 - Define the fundamental characteristics of an expert system.
 - What is the reason that an expert-system separates its knowledge base from the inference mechanism in the system?

Answer:

3. Given the rules below:

R1:	If X breathes using gills Then X lives in sea
R2:	If X has backbone and has fin and breathes using gills Then X is fish
R3:	If X lives in sea and has backbone and has fin Then X is seahorse
R4:	If X is fish and has fin and has two eyes at two sides Then X is shark
R5:	If X is fish and has two eyes at one side Then X is flounder

A subject with ID "X1001" was observed and some facts were gathered as listed below:

- ☐ X1001 has backbone
- ☐ X1001 has fins
- ☐ X1001 breathes through gills
- ☐ X1001's two eyes on two sides of its head

- From the observation above, conflicts may occur when you infer the rules above. Analyze what is the potential conflict that might occur. ()
- Suggest and discuss **THREE (3)** approaches of conflict resolution. ()
- Derive a conclusion for the problem above for each of the conflict resolution approaches that you have suggested in Question 3(b) by using forward chaining. ()

Answer:

Part II: Recommender Systems

1. There are two common types of recommendation engine algorithms, i.e. collaborative filtering models and content-based models. Briefly explain the differences between them. Provide examples of applications to aid your answers. ()

Answer:

2. Discuss the limitations of collaborative filtering models. ()

Answer:

3. Discuss the limitations of content-based models. ()

Answer:

4. There are two types of collaborative filtering models as follows. Differentiate these two types with the aid of suitable examples. ()
 - a) user-user
 - b) item-item

Answer:

Tutorial 12

Part I: Probability Theory

1. **FOUR (4)** of the rules in a rule-based system are given as follows:

R1: If there was drizzle last night, then there is 60% chance that the grass in the backyard is wet the next morning.
R2: If there was rain last night, then there is 80% chance that the grass in the backyard is wet the next morning.
R3: Drizzle occurs in 160 days in a year.
R4: Rain occurs in 120 days in a year.

Assume that the drizzle and rain occur exclusively, and there are 365 days in a year.

- a) Specify the above rules in probability assertion. ()
- b) If you see the grass in the backyard is wet in the morning, by using Bayes Theorem, what is the event that most probably occurred last night? ()
- c) After a conclusion is drawn from Question 1(b), your grandfather asserts that sometimes your neighbour would spray water in the backyard at night. ()
 - i. Do you think this assertion would affect the conditional probability?
 - ii. Do you think this assertion would affect your initial conclusion as well?

Explain your answers.

Answer:

2. Part of the rules in a liver disease diagnosis system is shown below. ()

R1: IF patient has loss of appetite
THEN patient has liver disease

R2: IF patient is a alcoholic
AND patient is a smoker
THEN patient has liver disease

Considering only the following information is given to you.

- i. $P(\text{patient has loss appetite} \mid \text{patient has liver disease}) = 0.8$
- ii. $P(\text{patient has liver disease}) = 0.00013$

Assume that a patient rated that he has loss appetite 70% of the time, demonstrate how do you predict $P(\text{patient has liver disease} \mid \text{patient has loss appetite})$ using Naïve Bayes Algorithm.

Answer:

one

$$0.7 \times 0.5 = 0.35$$

$$\text{mm}(0.8, 1.0) \times 0.9 = 0.72$$

$$0.35 + 0.75 - (0.35 \times 0.75)$$

$$= 0.818 \text{ almost certainly have lung disease}$$

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3. A police officer sees you run a red light late one Saturday night. Before pulling you over, the officer tries to guess what caused you to run the red light and comes up with the following possibilities:
- You are drunk
 - You are in a hurry
 - You are not wearing your glasses
 - You felt you could do it and get away with it

The policeman assigns the following evidential probabilities to these hypotheses given that the incident happened late on a Saturday night

- $P(\text{run light late Sat. night} \mid \text{drunk driver}) = 0.45$
- $P(\text{run light late Sat. night} \mid \text{driver in a hurry}) = 0.60$
- $P(\text{run light late Sat. night} \mid \text{driver didn't see light}) = 0.15$
- $P(\text{run light late Sat. night} \mid \text{didn't see cop/thinks can get away with it}) = 0.05$

He further assigns the following prior probabilities:

- $P(\text{drunk driver}) = 0.10$
- $P(\text{hurried driver}) = 0.33$
- $P(\text{driver not wearing glasses}) = 0.10$
- $P(\text{driver feels he can get away with things}) = 0.25$

Compute the conditional probabilities for the four possibilities. What decision is the police officer most likely to make? ()

Answer:

Part II: Certainty Factor

4. You receive a phone call from your classmate, James, informing you that your name is in the pass list. Preparing to tell your parents the good news, you recall that James is a liar who loves to play jokes on others. Thus, you decide to call another classmate, Sean. Although Sean is known for being occasionally careless, you believe that Sean is far more reliable. ()

R1: If James' call is true, then there is a 0.5 certainty that your name is in pass list.

R2: If Sean's call is true and Sean is not careless is true, then there is a 0.9 certainty that your name is in the pass list.

R3: if your name is in the pass list, then there is a 0.99 certainty that you pass the examination.

Conditions:

The certainty of James' call is 1.0.

The certainty of Sean's call is 1.0.

The certainty of Sean is not careless is 0.5

Based on the information provided, would you pass the examination?

Answer:

Part II: Certainty Factor

4. You receive a phone list. Preparing to tell jokes on others. The being occasionally c

(1.0)

R1: If James' call is true
R2: If Sean's call is true
name is in the pass list.
R3: if your name is in the

Conditions:
The certainty of James'
The certainty of Sean's c
The certainty of Sean is

Based on the inform

Answer:

Handwritten calculations on the left side of the slide:

R1: $1 \times 0.5 = 0.5$ pass list
R2: $\min(1, 0.5) \times 0.9 = 0.45$ pass list
R3: $CF_1 + CF_2 - (CF_1 \times CF_2)$
 $0.5 + 0.45 - (0.5 \times 0.45)$
 $= 0.725$ pass list
 $0.725 + 0.99$
 $= 0.7$ pass list
"prob..."

Handwritten answer: 0.005

5. Assume that a simple inference system is built using certainty factors. Demonstrate how to compute the certainty factor that a patient has liver disease based on the following rules. ()

R1:	IF	patient has loss of appetite [CF 0.7]
	THEN	patient has liver disease [CF 0.5]
R2:	IF	patient is an alcoholic [CF 0.8]
	AND	patient is a smoker [CF 1.0]
	THEN	patient has liver disease [CF 0.9]

Combining Certainty Factors:

$$\begin{array}{ll}
 CF1 + CF2 - CF1 * CF2 & \text{if both are positive,} \\
 CF1 + CF2 + CF1 * CF2 & \text{if both are negative,} \\
 \frac{CF1 + CF2}{1 - \min(|CF1|, |CF2|)} & \text{otherwise}
 \end{array}$$

Answer:

6. The Ebola outbreak has raised concerns of international spread in 2014, similar to the SARS epidemic in more than 10 years ago. Assume that you are assisting a medical centre to develop an online expert system that consults people on recognizing the symptoms of these diseases. Part of the rules of the system is as follows: ()

R1:	IF	body temperature exceeds 37 °C
	AND	headache presents AND muscle aches presents
	THEN	fever presents (CF 0.8)
R2:	IF	body temperature exceeds 37 °C
	THEN	fever presents (CF 0.2)
R3:	IF	body temperature exceeds 38 °C
	THEN	fever presents (CF 1.0)
R4:	IF	cough presents AND fever presents
	THEN	symptom of SARS (CF 0.5)
R5:	IF	cough presents AND cough starts 2-3 days after other symptoms
	AND	fever presents
	THEN	symptom of SARS (CF 0.6)
R6:	IF	IF (cough presents OR nausea presents OR stomach pain presents)
	AND	fever presents
	THEN	symptom of Ebola (CF 0.5)

R7:	IF	IF cough presents AND nausea presents AND stomach pain presents
	AND	fever presents
	THEN	symptom of Ebola (CF 0.8)

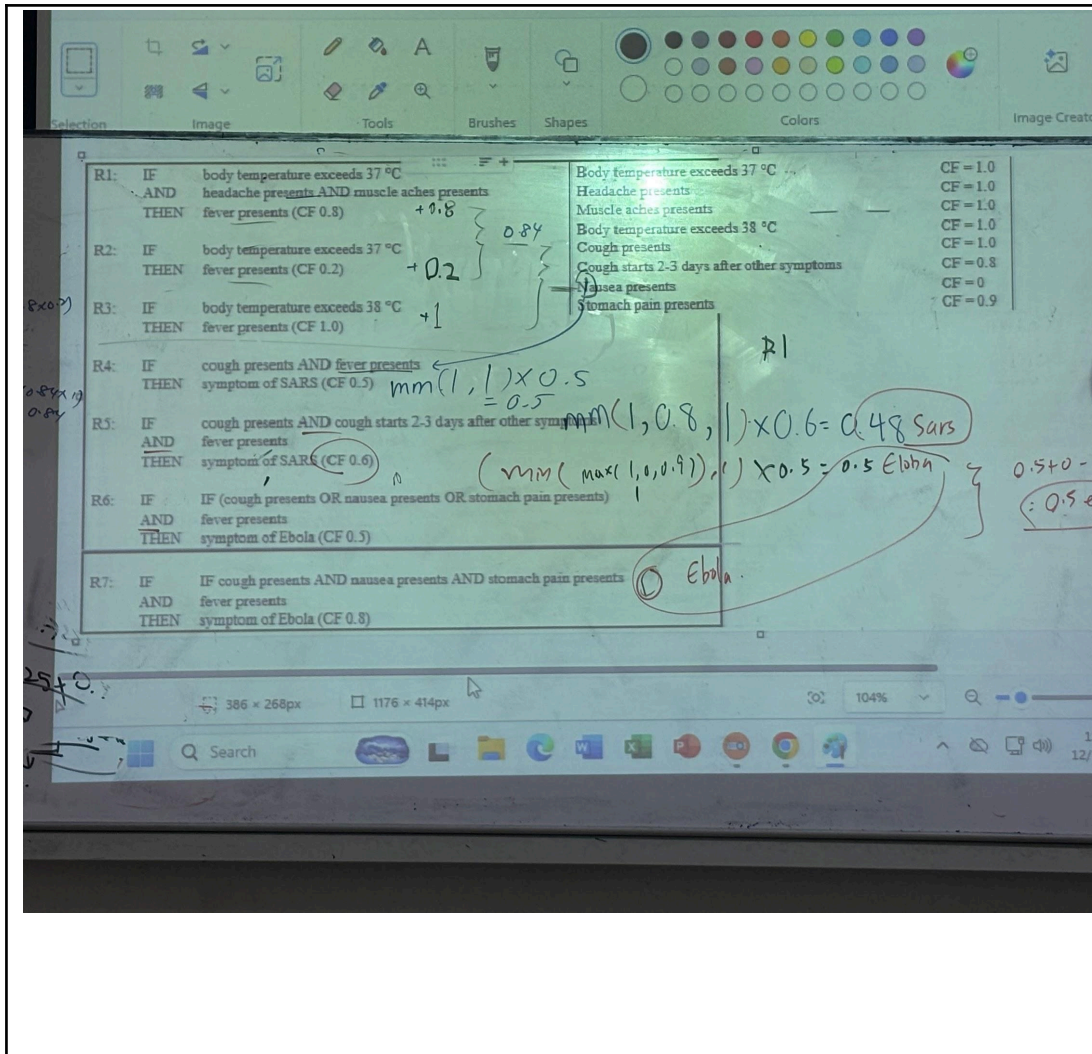
John has recently returned from overseas and he started to show the following symptoms. Based on R1 to R7, demonstrate the inference steps and conclude what disease that John could be infected.

Body temperature exceeds 37 °C	CF = 1.0
Headache presents	CF = 1.0
Muscle aches presents	CF = 1.0
Body temperature exceeds 38 °C	CF = 1.0
Cough presents	CF = 1.0
Cough starts 2-3 days after other symptoms	CF = 0.8
Nausea presents	CF = 0
Stomach pain presents	CF = 0.9

Remark: To calculate a combined certainty factor, use the following equation:

$$\begin{array}{ll} CF1 + CF2 - CF1*CF2 & \text{if both are positive,} \\ CF1 + CF2 + CF1*CF2 & \text{if both are negative,} \\ \frac{CF1 + CF2}{1 - \min(|CF1|, |CF2|)} & \text{otherwise} \end{array}$$

Answer:



7. Given the report of an observation at 7am as below, demonstrate the inference and conclude the weather of tomorrow based on certainty factors. ()

Place of observation: Miami, Florida (tropical climate) (CF=1)

Observation data:

- Cloud height > 18,000 feet (CF = 0.8)
- Clouds are small, rounded puffs, appeared in long rows. (CF=0.8)
- Cloud patterns are sheet-like. (CF = 0.2)
- Color of cloud is white. (CF=0.7)
- Sun can shine through. (CF=0.9)
- Humidity: warm and sticky (CF=0.7)

Remark: The certain factor (CF) for other unknown data is 0.

IF Cloud_Height > 18,000 feet [P1.1]
THEN Cloud_Group is Cirrus [H1] (CF= 0.8)

IF Cloud_Height is between 6,500 feet to 18,000 feet [P2.1]
THEN Cloud_Group is Alto [H2] (CF=0.8)

IF Cloud_Group is Cirrus [P3.1]
AND Cloud_Pattern is sheet-like [P3.2]
AND Cloud_Color is white [P3.3]
AND Moon or Sun can shine through [P3.4]

THEN Cloud_Type is Cirrostratus [H3] (CF=0.8)

IF Cloud_Group is Cirrus [P4.1]

AND Cloud_Pattern is small, rounded puffs with long rows [P4.2]

AND Cloud_Color is white or gray [P4.3]

THEN Cloud_Type is Cirrocumulus [H4] (CF=0.8)

IF Cloud_Group is Alto [P5.1]

AND Cloud_Color is gray or blue gray [P5.2]

AND Moon or Sun can shine through [P5.3]

THEN Cloud_Type is Altostratus [H5] (CF=0.7)

IF Cloud_Group is Alto [P6.1]

AND Cloud_Color is gray [P6.2]

THEN Cloud_Type is Altocumulus [H6] (CF=0.8)

IF Cloud_Type is Cirrostratus [P7.1]

THEN Rain or snow storm occurs within 12-24 hours [H7] (CF=0.7)

IF Cloud_Type is Cirrocumulus [P8.1]

AND Climate_Zone is NOT of Tropical [P8.2]

THEN Weather is fair but cold [H8] (CF=0.7)

IF Cloud_Type is Cirrocumulus [P9.1]

AND Climate_Zone is of Tropical [P9.2]

THEN Hurricane is approaching [H9] (CF=0.6)

IF Cloud_Type is Altostratus [P10.1]

THEN Storms with continuous rain or snow is reaching [H10] (CF=0.7)

IF Cloud_Type is Altocumulus [P11.1]

AND Morning is warm and sticky [P11.2]

THEN Thunderstorms by late afternoon [H11] (CF=0.7)

Part of the rules in the expert system (Reference: boatsafe.com)

Answer:



Tutorial 13

1. Following shows a sample crisp rule. ()

IF temperature > 37.5 THEN fever = TRUE
--

- a) Explain the difference between crisp rule and fuzzy rule.
- b) Suggest how do you convert the crisp rule above into a fuzzy rule.
- c) Design the appropriate fuzzy sets to represent the input and output variables respectively.

Answer:

2. Illustrate the general structure of Fuzzy Inference System and explain each of the components. ()

Answer:

3. Assume that you are designing an automated air conditioning system using Fuzzy Logic. ()
- a) Suggest an uncertainty in the above scenario with explanation.

- b) Generate the **TWO (2)** input and **TWO (2)** output fuzzy membership functions that are able to model the air conditioning system.
- c) Suggest **TWO (2)** fuzzy rules that associate to the Fuzzy membership functions generated in Question 3(b).

Answer:

4. Suppose you are designing a fuzzy thermostat with the following membership function of the temperature. ()

Low (x) = $\begin{cases} 0, & \text{if temp (x) } \geq 21, \\ f(\text{temp (x)}), & \text{if } 19 < \text{temp (x)} < 21, \\ 1, & \text{if temp (x) } \leq 19 \end{cases}$

Medium (x) = $\begin{cases} 0, & \text{if temp (x) } \leq 20 \text{ or temp (x) } \geq 28, \\ f(\text{temp (x)}), & \text{if } 20 < \text{temp (x)} < 23 \text{ or } 25 < \text{temp (x)} < 28 \\ 1, & \text{if } 23 \leq \text{temp (x)} \leq 25 \end{cases}$

High (x) = $\begin{cases} 0, & \text{if temp (x) } \leq 27, \\ f(\text{temp (x)}), & \text{if } 27 < \text{temp (x)} < 30 \\ 1, & \text{if temp (x) } \geq 30 \end{cases}$

- (i) Illustrate the graph with labels to represent the above membership function.
- (ii) Propose **ONE (1)** fuzzy rule that can be used in the fuzzy thermostat.

Answer: